

Low-energy Earth-Moon transfers with lunar ballistic capture based on Lagrangian coherent structures in a four-body model

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Abstract

This paper investigates lunar ballistic capture and the construction of low-energy bi-impulsive Earth-Moon transfers, focusing on the effects of energy levels and Sun-perturbed dynamics, utilizing Lagrangian coherent structures (LCSs) for low Moon insertion impulse. The bi-impulsive Earth-Moon transfers are established in the context of the Sun-Earth/Moon planar bicircular restricted four-body problem (PBCR4BP), categorized into interior and exterior transfers according to the positions of the Poincaré sections. First, initial guesses for interior/exterior transfers are generated from the initial states inside the LCSs. Based on the initial guesses, arrival-in performance influencing the Moon insertion impulse are discussed to reveal the dependence law. According to the arrival-in performance, all the tested transfers end up in lunar ballistic capture. The ballistic capture is further categorized into direct and retrograde capture. Then, the construction strategy for LCS-based low-energy transfers is proposed based on the dependence law. Initial guesses with direct/retrograde capture are selected based on the construction strategy and corrected to obtain the LCS-based transfers, resulting in Moon insertion impulse solutions (0.628–0.640 km/s) closer to the theoretical minimum value (0.625 km/s). Families of low-energy transfers in four scenarios (i.e., interior/exterior transfers with direct/retrograde capture) are investigated without direct optimization. Low-energy transfers are achieved in a more targeted way, and typical trajectories reveal the role of solar gravity perturbation, natural dynamical structure LCS and lunar ballistic capture in the transfer dynamics.

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Keywords: Low-energy Earth-Moon transfer; Lunar ballistic capture; Planar bicircular restricted four-body problem; Lagrangian coherent structure

1. Introduction

Renewed interest in lunar exploration has sparked worldwide with the proposal and implementation of various missions (e.g., *Artemis I* (Batcha et al., 2020), *Chang'e-5* (Zheng et al., 2023), *Chandrayaan-2* (Renju and Raju, 2023)), deepening mankind's scientific explo-

ration of the cislunar space and the Moon. compared to other mission scenarios (e.g., libration point transfer missions (Du et al., 2023)), the Earth-Moon transfer mission is important for assessing the lunar gravitational field (e.g., *GRAIL* (Hatch et al., 2010) and *Danuri* (Song et al., 2023)), constructing lunar infrastructure (e.g., *Lunar Scientific Research Base* (Pollpeter et al., 2023)), and supports human lunar exploration. Therefore, how to utilize the theory and knowledge of astrodynamics to construct Earth-Moon transfers has been widely investigated. Moreover,

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the low-energy Earth-Moon transfer, characterized by low fuel consumption and flexible launch window (Parker and Anderson, 2014), has been widely favored in practical missions (e.g., *Hiten* (Belbruno and Miller, 1993)). The impulsive Earth-Moon transfer can usually be simplified into a bi-impulsive trajectory during the preliminary design stage (Pan et al., 2022). After conducting an Earth injection impulse, a spacecraft is launched to a transfer trajectory from an Earth parking orbit and inserted into a target circumlunar orbit through a Moon insertion impulse (Topputo, 2013; Oshima et al., 2019). In the bi-impulsive transfer scenario, the magnitude of the Earth injection impulse is mainly determined by the capability of the launch vehicle engines (Mingotti et al., 2012), while the Moon insertion impulse primarily depends on the fuel capacity of the spacecraft engines. Therefore, reducing the Moon insertion impulse during the transfer process can lead to a reduction in the fuel requirements of spacecraft and allow for an increase in the payload mass. For this consideration, this paper focuses on the distribution and influencing factors of Moon insertion impulses (i.e., arrival-in performances) in the bi-impulsive Earth-Moon transfers and the construction of low-energy transfers with low Moon insertion impulses.

The selection of the dynamical model for transfer trajectory design is crucial for reducing fuel consumption. The patched conic method based on the patched restricted two-body problem (R2BP) (Crenshaw, 1963; Battin, 1999) was first used in constructing transfer trajectories. This approach yielded solutions with short times of flight (TOF) but incurred high total impulses due to the lack of application of multi-body gravity. Addressing this issue, Sweetser (1991) developed a theoretical framework for estimating the minimum total impulse required for a bi-impulsive Earth-Moon transfer based on the Earth-Moon circular restricted three-body problem (CR3BP). This work highlighted the potential to reduce both the total and Moon insertion impulse by using multi-body gravity. Belbruno and Miller (1993) pioneered the application of the weak stability boundary (WSB) theory in a multi-body model to the design of low-energy transfers, and the low-energy solution was validated in the *Lunar Observer* and *Hiten* mission. In their work, the Moon insertion impulse of the transfer trajectory was effectively reduced due to lunar ballistic capture (LBC). Meanwhile, the invariant manifold theory in the CR3BP was developed (?). Koon et al. (2001) developed low-energy Earth-Moon transfers by patching the invariant manifolds in the Sun-Earth/Moon coupled CR3BP, achieving a significant reduction in total and Moon insertion impulses compared to the results of the patched conic method. This method demonstrated the potential of dynamical structures in achieving LBC and reducing both the total and the Moon insertion impulses. Based on the aforementioned work, Ren and Shan (2014) conducted a detailed analysis of the variation in Moon insertion impulse with Jacobi energy by analyzing the transit orbits and invariant manifolds in

the Earth-Moon CR3BP. They concluded that the Moon insertion impulse is negatively correlated to Jacobi energy. Furthermore, Topputo (2013) and Oshima et al. (2019) demonstrated that utilizing the Sun-Earth/Moon bicircular restricted four-body problem (BCR4BP) to design transfer trajectories can achieve a further decrease in the total impulse. Consequentially, there is a potential for further reducing the Moon insertion impulse based on the BCR4BP. Motivated by this potential, we adopt the Sun-Earth/Moon planar BCR4BP (PBCR4BP) to investigate the arrival-in performances and the construction of low-energy transfers with low Moon insertion impulses. However, the characteristics of arrival-in performances in the PBCR4BP and the influencing factors (e.g., energy, perturbation) have not been thoroughly discussed. Additionally, conventional methods (Yagasaki, 2004; Topputo, 2013; Oshima et al., 2019) primarily use direct optimization and global search to construct Earth-Moon transfer trajectories in the PBCR4BP. These methods can be further improved by effectively using prior knowledge about impulse and initial state distributions. Therefore, this paper is devoted to obtaining priori knowledge about arrival-in performances and proposing a strategy for constructing low-energy transfers in a more targeted way in the PBCR4BP.

Firstly, we focus on obtaining priori knowledge about arrival-in performances. We note that the Moon insertion impulse is typically closely associated with the transit orbits, which are the trajectories that pass through the Earth-Moon L1/L2 region and form an integral part of the entire transfer process (Ren and Shan, 2014). In the CR3BP, the invariant manifold (?; Koon et al., 2006) is the dynamical structure that separates transit and non-transit orbits. However, in the BCR4BP, the time-dependent dynamics precludes the direct application of invariant manifolds (Onozaki and Yoshimura, 2013). To address this issue, in this paper, the Lagrangian coherent structure (LCS) (Short and Howell, 2014) is introduced to separate transit and non-transit orbits and construct bi-impulsive transfers (i.e., LCS-based transfers). LCSs are defined as the ridges in the finite time Lyapunov exponent (FTLE) fields (Onozaki et al., 2017). Initial states inside the LCSs can generate transit orbits. Therefore, the initial state set that can generate transfer trajectories is a subset of the initial state set inside the LCSs. From this, it can be concluded that the characteristics of LCS-based transfers, including arrival-in performances, are closely associated with the characteristics of the LCSs. According to Qi et al. (2012) and Fu et al. (2024), we know that the configurations of LCSs in the PBCR4BP vary with energy levels and Sun-perturbed dynamics. Therefore, to evaluate the influence of energy levels and Sun-perturbed dynamics on the Moon insertion impulses, we parameterize the LCSs by the energy levels and solar phase angles, performing a global search of the initial states inside the LCSs to generate initial guesses of transfers. Since the way of LBC (i.e., direct and retrograde capture) has a further influence on

the arrival-in performances (Oshima et al., 2019), the influence of the angular momentum with respect to the Moon is also analyzed. As a result, we discover some phenomena that have not yet been reported. It is found that retrograde capture requires higher energy levels compared to direct capture. The presence of Sun-perturbed dynamics influences the energy conditions for LBC. Then, the dependence law of the Moon insertion impulse and priori knowledge for narrowing the search space for low-energy transfers are summarized based on the analysis of arrival-in performances. According to the LCSs associated with the L1/L2 region (i.e., interior/exterior-LCSs) and direct/retrograde LBC, we categorize the Earth-Moon transfers into four scenarios: interior transfer with direct capture, interior transfer with retrograde capture, exterior transfer with direct capture, and exterior transfer with retrograde capture. For these scenarios, the construction strategy of low-energy transfers with low Moon insertion impulse is proposed based on the dependence law and priori knowledge. Simulations illustrate that our proposed strategy can effectively construct low-energy transfers, and comparisons with solutions obtained from other dynamical models demonstrate that Sun-perturbed dynamics helps reduce the fuel consumption of Earth-Moon transfers. Above all, the contributions of our work are listed as follows:

- (1) New phenomena regarding arrival performances have been discovered. Prior knowledge about transfer dynamics in the PBCR4BP is summarized to construct low-energy transfers in a targeted way.
- (2) Families of low-energy transfers with low total and Moon insertion impulses for proposed four scenarios (i.e., interior/exterior transfers with direct/retrograde capture) are investigated, which confirm and extend the conclusions of Campagnola and Russell (2010), Ren and Shan (2014) and Oshima et al. (2019).

The rest of this paper is organized as follows. The LCS in the PBCR4BP are detailed in Section 2. Section 3 presents the construction process of LCS-based bi-impulsive transfers, summarizing priori knowledge about low-energy transfers and arrival-in performances. Section 4 discusses the results of LCS-based transfers. The characteristics of developed transfers are analyzed to reveal the role of the solar gravity perturbation, LCS, and LBC in reducing total and Moon insertion impulse. Finally, conclusions are drawn in Section 5.

2. Lagrangian coherent structures in the planar bicircular restricted four-body problem

Lagrangian coherent structures serve as almost invariant set in the PBCR4BP, which can generate transit orbits and construct low-energy transfers. In this section, the concepts of the Sun-Earth/Moon PBCR4BP, LCS, and the existence conditions of the LCS are briefly introduced.

2.1. Planar bicircular restricted four-body problem

The Sun-Earth/Moon planar bicircular restricted four-body problem (PBCR4BP) is adopted in this paper in which all primaries are assumed to move on the same plane. The PBCR4BP involves four primaries, denoted by P_1, P_2, P_3 , and P_4 . P_4 is regarded as a massless particle and moves in the multi-gravitational fields of P_1, P_2 , and P_3 . In this paper, P_1, P_2 , and P_3 indicates the Sun, the Earth and the Moon, respectively, and P_4 is the spacecraft. The Sun-Earth/Moon PBCR4BP can be considered as a simplified model of the real Sun-Earth/Moon system, and more studies on transfer trajectories in the spatial Sun-Earth/Moon system can be found in the work from Gong et al. (2007), and transfers in the elliptic restricted four-body problem can be found in the work from Liu and Gong (2018). As shown in Fig. 1, a rotating frame ($O-x-y-z$) is established, where the origin is at the barycenter of the Earth-Moon system. The x -axis is directed from the Earth to the Moon and the z -axis is directed to the angular momentum vector of the Earth-Moon system. The y -axis is determined by the right-hand rule. In this paper, we only focus on the motions within the (x, y) plane, as shown in Fig. 1.

The dimensionless units are set as follows: the length unit (LU) indicates the Earth-Moon distance; the time unit (TU) is set as $TU = T_{EM}/2\pi$, where T_{EM} is the period of the Earth and the Moon orbiting around their barycenter. Then, the normalized dynamical equations of spacecraft in the PBCR4BP can be written as

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{u} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} u \\ v \\ 2v + \frac{\partial \Omega_4}{\partial x} \\ -2u + \frac{\partial \Omega_4}{\partial y} \end{bmatrix} \quad (1)$$

$$\Omega_4 = \frac{1}{2} [x^2 + y^2 + \mu(1 - \mu)] + \frac{1 - \mu}{r_1} + \frac{\mu}{r_2} + \frac{\mu_S}{r_3} - \frac{\mu_S}{\rho^2} (x \cos \theta_S + y \sin \theta_S) \quad (2)$$

where $r_1 = \sqrt{(x + \mu)^2 + y^2}$ and $r_2 = \sqrt{(x + \mu - 1)^2 + y^2}$, indicating the distances from the spacecraft to the Earth and the Moon. $\mathbf{X} = [x, y, u, v]^T$ is the orbital state, Ω_4 is the effective potential of the PBCR4BP, $\mu = m_M/(m_E + m_M)$ is the Earth-Moon mass parameter, and $\mu_S = m_S/(m_E + m_M)$ is the dimensionless mass of the Sun. The masses of the Sun, the Earth, and the Moon are denoted by m_S, m_E , and m_M , respectively. The solar gravity perturbation (SGP) depends on the distance between the Sun and the Earth-Moon barycenter ρ , the distance between the spacecraft and the Sun $r_3 = \sqrt{(x - \rho \cos \theta_S)^2 + (y - \rho \sin \theta_S)^2}$, and the solar phase angle θ_S . Moreover, θ_S satisfies

$$\theta_S = \theta_{S0} + \omega_S T \quad (3)$$

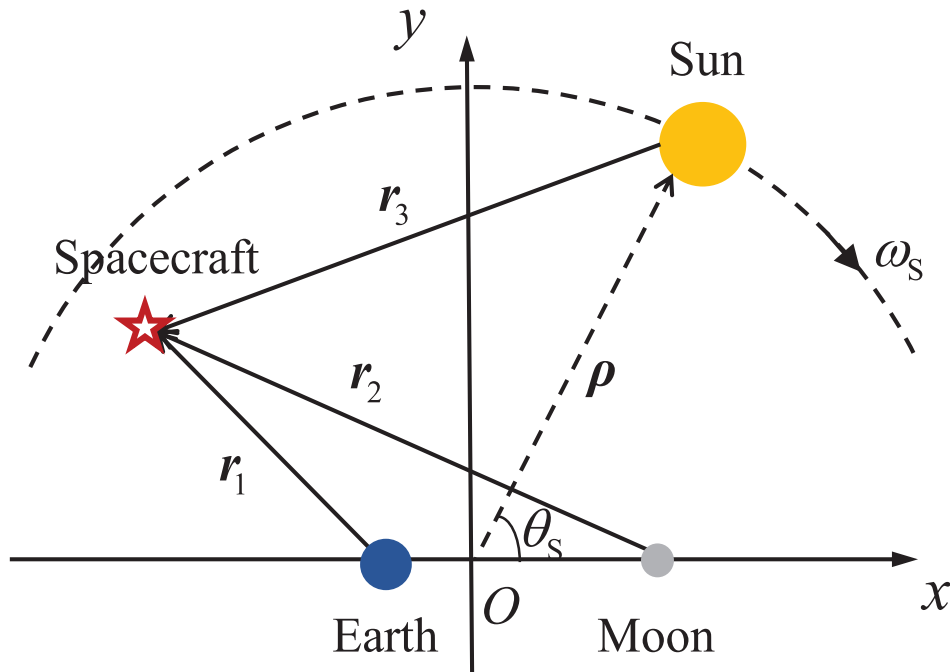


Fig. 1. PBCR4BP in the Earth-Moon rotating frame.

where θ_{s0} is the phase angle at the initial epoch t_0 and ω_S is the synodic angular velocity of the Sun with respect to the Earth-Moon system. In particular, at $t_0 = 0$, the Sun is placed on the x -axis, corresponding to a lunar eclipse. $T = t - t_0$ indicates the integration time up to the epoch t . The specific values of the aforementioned physical constants used in the simulation are summarized in Table 1.

The Hamiltonian indicating the generalized energy is defined as (Qi et al., 2012)

$$H = \frac{1}{2}(u^2 + v^2) - \frac{1}{2}(x^2 + y^2) - \frac{1-\mu}{r_1} - \frac{\mu}{r_2} - \frac{\mu_S}{r_3} + \frac{\mu_S}{\rho^2}x \times \cos \theta_S + \frac{\mu_S}{\rho^2}y \sin \theta_S - \frac{1}{2}\mu(1-\mu) \quad (4)$$

A higher Hamiltonian level corresponds to increased energy of the spacecraft, which is time-dependent due to the solar gravitational perturbation (SGP). Following the

concept of Hill region (Koon et al., 2006) the instantaneous Hill region is defined by the initial Hamiltonian H_0 and the initial solar phase angle θ_{s0} at the initial epoch t_0 (i.e., $\theta_{s0} = \omega_S t_0$) (Qi et al., 2012)

$$M_0(H_0, \theta_{s0}) = \{(x, y) | H(x, y, u, v, \theta_{s0}) \leq H_0, u = v = 0\} \quad (5)$$

Therefore, given a specific initial Hamiltonian H_0 and initial solar phase angle θ_{s0} , the reachable region can be mainly categorized into the Earth region, the Moon region, the L1 region, the L2 region, and the Earth-Moon exterior region (Yamato and Spencer, 2004), as shown in Fig. 2. Additionally, the H_0 value required for the L2 region to exist is higher than that for the L1 region. Due to the time-dependence of the PBCR4BP, the trajectory may insert into the instantaneous forbidden region determined by H_0 and θ_{s0} (black closed curves in Fig. 2). In this paper,

Table 1
Physical constants of the PBCR4BP (Topputo, 2013).

Symbol	Value	Units	Meaning
μ	$1.21506683 \times 10^{-2}$	–	Earth-Moon mass parameter
μ_S	3.28900541×10^5	–	Dimensionless mass of the Sun
ρ	3.88811143×10^2	–	Dimensionless Sun-Earth/Moon distance
ω_S	$-9.25195985 \times 10^{-1}$	–	Synodic angular velocity
T_{EM}	2.24735067×10^6	s	Earth-Moon period
R_E	6378.145	km	Mean Earth's radius
R_M	1737.100	km	Mean Moon's radius
LU	3.84405000×10^5	km	Length unit
TU	3.75676968×10^5	s	Time unit

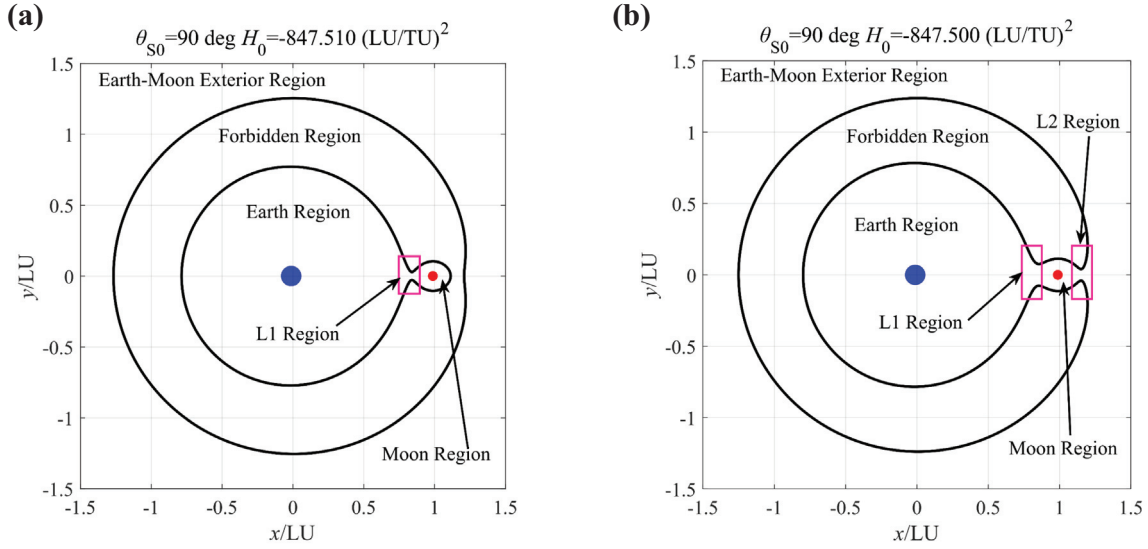


Fig. 2. Configuration of instantaneous Hill region. (a) Instantaneous Hill region with L1 region (the H_0 value is low enough, causing the L2 region to remain closed). (b) Instantaneous Hill region with L1 region and L2 region.

we define the interior/exterior Earth-Moon transfer as transfer trajectories finally encountering the Moon through the L1/L2 region (Oshima et al., 2019). Moreover, the transfer trajectories are constructed from Lagrangian coherent structures (LCSs) in the PBCR4BP.

2.2. Lagrangian coherent structures and existence conditions

Similar to the invariant manifolds of the autonomous circular restricted three-body problem (CR3BP), a dynamical structure named LCS is introduced to describe the phase space transport and generate transit orbits (Short and Howell, 2014) of the non-autonomous system, i.e., the PBCR4BP in this paper. The LCSs are defined as the ridges in the finite time Lyapunov exponent (FTLE) fields (Onozaki et al., 2017). To reduce the dimension of the phase space, a Poincaré section visualizing the FTLE fields and LCSs is introduced as (Qi et al., 2012)

$$U = \{(y_0, v_0) | x_0 = \text{const}, H_0 = \text{const}, \theta_{S0} = \text{const}\} \quad (6)$$

In the following texts, we use the subscript "0" to represent the quantities associated with the selected Poincaré section. Based on the Poincaré section, $u_0 = \dot{x}_0$ is calculated as

$$u_0 = \pm \sqrt{-2H_0 + x_0^2 + y_0^2 - v_0^2 + \frac{2(1-\mu)}{r_{10}} + \frac{2\mu}{r_{20}} + \frac{2\mu_S}{r_{30}} - \frac{2\mu_S}{\rho^2} (x_0 \cos \theta_{S0} + y_0 \sin \theta_{S0}) + \mu(1-\mu)} \quad (7)$$

where (y_0, v_0) is selected from the grid on the Poincaré section and x_0 is given. The positive sign of u_0 indicates a trajectory towards the Moon, whereas the negative u_0 generates a trajectory towards the Earth (Short and Howell, 2014). In this paper, we set its sign positive to construct trajectories to the Moon.

After the introduction of the Poincaré section, the five-dimensional phase space (together with time) is reduced to (y_0, v_0) plane. To visualize the LCSs, the FTLE fields

are projected onto (y_0, v_0) plane by calculating the value of FTLE at each point as

$$\sigma(X_0, t_0, T) = \frac{1}{T} \ln \left(\sqrt{\lambda_{\max} \left(\left(\nabla \phi_{t_0}^{t_0+T}(X_0) \right)^T \nabla \phi_{t_0}^{t_0+T}(X_0) \right)} \right) \quad (8)$$

where $\lambda_{\max} \left(\left(\nabla \phi_{t_0}^{t_0+T}(X_0) \right)^T \nabla \phi_{t_0}^{t_0+T}(X_0) \right)$ is the maximum eigenvalue of the Cauchy-Green tensor, $C = \left(\nabla \phi_{t_0}^{t_0+T}(X_0) \right)^T \nabla \phi_{t_0}^{t_0+T}(X_0)$, and $\phi_{t_0}^{t_0+T} : \mathbb{R} \times \mathbb{R} \times \mathbb{D} \rightarrow \mathbb{D}; (t_0, t_0 + T, X_0) \rightarrow \phi_{t_0}^{t_0+T}(X_0)$ is the flow map of the PBCR4BP. t_0 is the initial epoch at the Poincaré section and $T = t - t_0$ indicates the integration time up to the epoch t . Lagrangian coherent structures are the ridges in the FTLE fields and serve as almost invariant sets in the PBCR4BP (i.e., separating the transit orbits passing through the L1/L2 region from those non-transit orbits) (Qi et al., 2012). Here, we give two examples to visualize the LCSs, the Poincaré section is set as

$$U_0 = \{(y_0, v_0) | x_0 = 0.5\text{LU}, H_0 = -847.500(\text{LU/TU})^2, \theta_{S0} = 90 \text{ deg}\} \quad (9)$$

Given that $T = 7\text{TU}$, the FTLE fields associated with the interior-LCS and exterior-LCS are shown in Fig. 3 and Fig. 4, respectively. In the figures, there are ridges (i.e., black closed curve) in the FTLE field, defined as LCSs existing in the FTLE fields. The initial states inside the LCSs generate transit orbits through forward-time propagation, whereas those outside result in non-transit orbits (see patterns of transit and non-transit orbits in Figs. 3,4). Similar to the invariant manifolds of L1/L2 Lyapunov orbits in the CR3BP, the LCSs can be categorized into interior-LCSs (i.e., generating the transit orbits passing

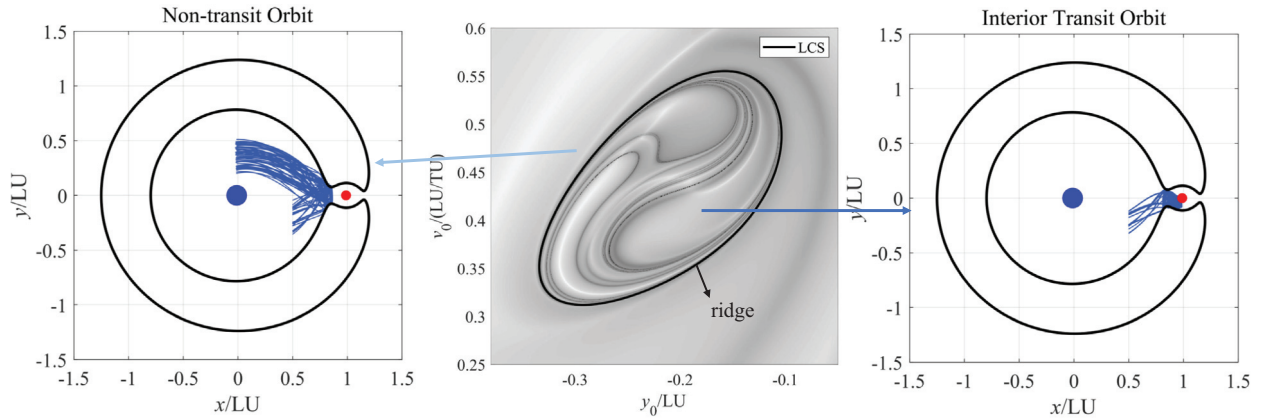


Fig. 3. FTLE field and interior-LCS on U_0 section. Initial states inside the LCS generate the transit orbits passing through the L1 region and reaching the Moon region, while those outside generate non-transit orbits bouncing back to the Earth region.

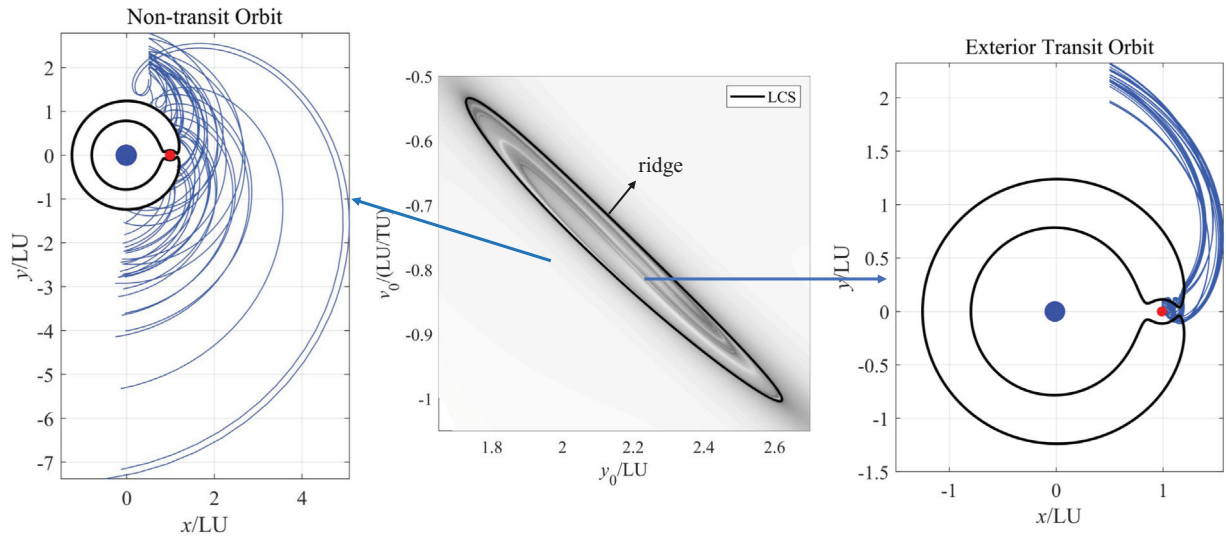


Fig. 4. FTLE field and exterior-LCS on U_0 section. Initial states inside the LCS generate the transit orbits passing through the L2 region and reaching the Moon region, while those outside generate non-transit orbits bouncing back to the Earth-Moon exterior region.

through the L1 region) and exterior-LCSs (i.e., generating the transit orbits through the L2 region). In Fig. 3, the ridge represented by the black closed curve is classified as an interior-LCS, while the ridge in Fig. 4 is identified as an exterior-LCS. The interior-LCSs and exterior-LCSs are distributed in different positions of the Poincaré sections (i.e., $(y_0 < 0, v_0 > 0)$ for interior-LCSs and $(y_0 > 0, v_0 < 0)$ for exterior-LCSs investigated in this paper). Due to the separating properties, the LCSs bound the region on the map from which initial states are selected for constructing Earth-Moon transfer trajectories.

Following these examples, we assume that x_0 of the Poincaré section is set to 0.5 LU (i.e., the Poincaré section is set as $U_1 = \{(y_0, v_0) | x_0 = 0.5 \text{ LU}, H_0, \theta_{s0}\}$ and the integration time is set to $T = 7 \text{ TU}$ in the following texts. The assumed integration time ensures that transit orbits have enough time to reach the Moon region. Therefore, the

LCSs can be parameterized by (θ_{s0}, H_0) . Note that at a given θ_{s0} value, if the H_0 value of U_1 is low enough, the L1/L2 region and the associated LCS will vanish (see no ridges, i.e., no LCSs existing under low H_0 values in Fig. 5) (Qi et al., 2012). Therefore, there is a minimum value of H_0 (H_{0min}) required for both interior-LCSs and exterior-LCSs to exist at a given θ_{s0} value. The exact value of H_{0min} is numerically obtained and listed in Table 2 for interior-LCSs and exterior-LCSs, respectively. When the θ_{s0} values are varying, the configurations of LCSs translate along the H_0 axis (Qi et al., 2012). Moreover, the translation also exists in (y_0, v_0) plane (Fu et al., 2024). Consequently, the properties of the corresponding transit orbits can vary. This indicates that (θ_{s0}, H_0) parameters significantly influence the dynamical behavior of the LCSs, and consequentially affects the characteristics of the transfer trajectories generated from the LCSs. Subsequently, the

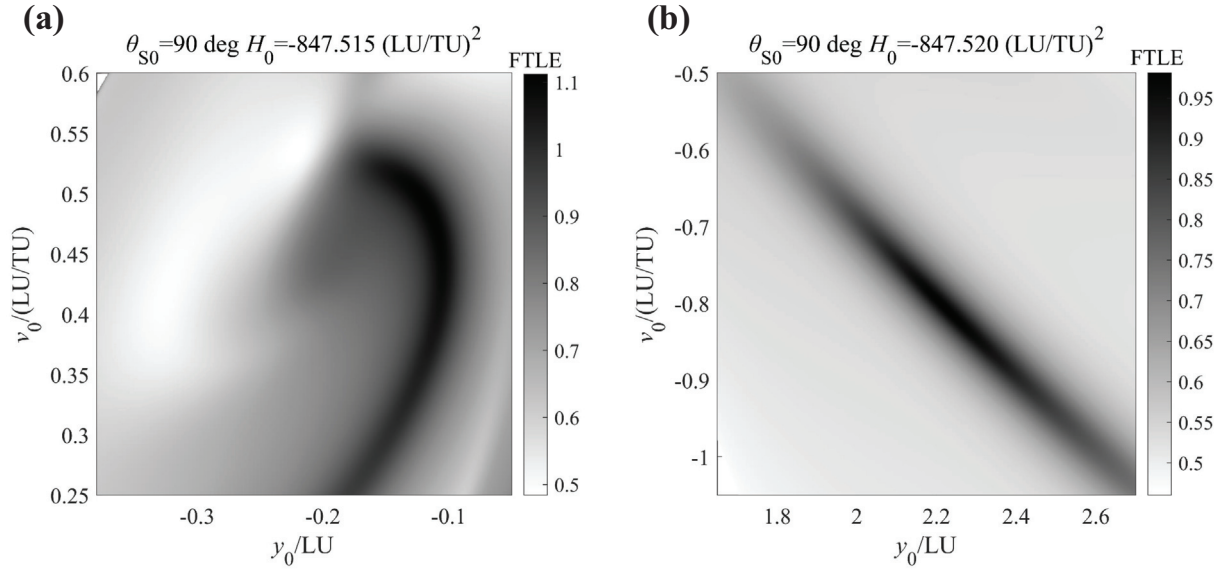


Fig. 5. FTLE fields without ridges (LCSs). (a) Interior circumstance. (b) Exterior circumstance.

Table 2
 $H_{0\min}/(\text{LU/TU})^2$ at each θ_{S0} for interior-LCSs and exterior-LCSs.

θ_{S0}/deg	0	90	180	270
Interior-LCS	−847.515	−847.510	−847.515	−847.510
Exterior-LCS	−847.495	−847.515	−847.495	−847.515

construction of bi-impulsive Earth-Moon transfers with low Moon insertion impulse based on the LCSs (i.e., bi-impulsive LCS-based transfers) is presented in Section 3.

3. Construction of bi-impulsive LCS-based transfers with low v_f

This section presents the construction process of the bi-impulsive LCS-based transfer. Firstly, the bi-impulsive LCS-based transfer is defined. Then, to construct bi-impulsive LCS-based transfers with low Δv_f , we focus on obtaining priori knowledge about arrival-in performances. We perform the LCSs under different (θ_{S0}, H_0) to generate initial guesses of transfers. Subsequently, arrival-in performance at the lunar insertion point (e.g., the Kepler energy with respect to the Moon (E_f), the angular momentum with respect to the Moon (M_f) and the Jacobi energy (C_f)) of initial guesses are analyzed to obtain the dependence law of Δv_f . Based on the law, we present the construction strategy for LCS-based transfers with low Δv_f . Finally, the trajectories are corrected from the selected initial guesses by solving the nonlinear programming (NLP) problem.

3.1. LCS-based bi-impulsive Earth-Moon transfer

In recent years, a variety of transfer scenarios in the cis-lunar space (e.g., libration point transfers (Du et al., 2024), impulsive Earth-Moon transfers (Topputo, 2013), and low-

thrust Earth-Moon transfers (Mingotti et al., 2012)) have been increasingly considered. Each scenario has different mission requirements. In particular, this paper focuses on one of aforementioned scenarios, i.e., the bi-impulsive Earth-Moon transfers. After conducting an Earth injection impulse (Δv_i), a spacecraft is launched to a transfer trajectory from a circular Earth parking orbit and inserted into a circular target circumlunar orbit through a Moon insertion impulse (Δv_f). Bi-impulsive Earth-Moon transfers can be mainly categorized into interior and exterior transfers, based on whether the spacecraft finally encounters the Moon through the L1 or L2 region (i.e., L1 for interior transfers and L2 for exterior transfers) (Oshima et al., 2019). Moreover, the impulses should be tangential to maximize spacecraft energy variations according to Pernicka et al. (1994).

The geometric constraints of the bi-impulsive Earth-Moon transfers are (Topputo, 2013):

$$\psi_i = \begin{bmatrix} (x_i + \mu)^2 + y_i^2 - (R_E + h_i)^2 \\ (x_i + \mu)(u_i - y_i) + y_i(v_i + x_i + \mu) \end{bmatrix} = \mathbf{0} \quad (10)$$

$$\psi_f = \begin{bmatrix} (x_f + \mu - 1)^2 + y_f^2 - (R_M + h_f)^2 \\ (x_f + \mu - 1)(u_f - y_f) + y_f(v_f + x_f + \mu - 1) \end{bmatrix} = \mathbf{0} \quad (11)$$

where $(x_i + \mu)(u_i - y_i) + y_i(v_i + x_i + \mu) = 0$ and $(x_f + \mu - 1)(u_f - y_f) + y_f(v_f + x_f + \mu - 1) = 0$ ensure the spacecraft departs from the Earth parking orbit/inserts into the target circumlunar orbit tangentially (Pernicka et al., 1994). The altitudes of the Earth parking orbit h_i and the target circumlunar orbit h_f are selected based on the mission requirements. The subscript “i” corresponds to the quantities associated with the Earth parking orbits, while the subscript “f” corresponds to those of the target

circumlunar orbits. Satisfying the constraints (i.e., Eqs. (10)), the Δv_i and Δv_f are calculated by (Topputo, 2013)

$$\Delta v_i = \sqrt{(u_i - y_i)^2 + (v_i + x_i + \mu)^2} - \sqrt{\frac{1 - \mu}{R_E + h_i}} \quad (12)$$

$$= \sqrt{(u_i - y_i)^2 + (v_i + x_i + \mu)^2} - \sqrt{\frac{1 - \mu}{\sqrt{(x_i + \mu)^2 + y_i^2}}}$$

$$\Delta v_f = \sqrt{(u_f - y_f)^2 + (v_f + x_f + \mu - 1)^2} - \sqrt{\frac{\mu}{R_M + h_f}} \quad (13)$$

$$= \sqrt{(u_f - y_f)^2 + (v_f + x_f + \mu - 1)^2} - \sqrt{\frac{\mu}{\sqrt{(x_f + \mu - 1)^2 + y_f^2}}}$$

Therefore, the total impulse (Δv) of bi-impulsive transfers is

$$\Delta v = \Delta v_i + \Delta v_f \quad (14)$$

The altitude of the Earth parking orbit is set to 36000 km (Cardona et al., 2016) and for the target circumlunar orbital altitude it is 100 km (Oshima et al., 2019). The altitude setting aids in analyzing bi-impulsive transfers from the Geosynchronous Earth Orbit (GEO) to the low Moon orbit (LMO). This transfer scenario could serve as an intermediate step for transfer to a lunar quasi-frozen orbit (Singh et al., 2019). Furthermore, the results presented in this paper, based on this altitude setting, can be extended to other orbital altitude settings with no significant qualitative difference expected. As shown in Fig. 6, a LCS-based bi-impulsive transfer trajectory consists of two segments, i.e., a transit segment (inserting into the target circumlunar orbit) and an Earth escape segment (departing from the Earth parking orbit). The transit segment is generated by forward-time propagation of the initial states inside the LCSs (i.e., transit orbits defined in Section 2.2). Conversely, the Earth escape segment results from backward-

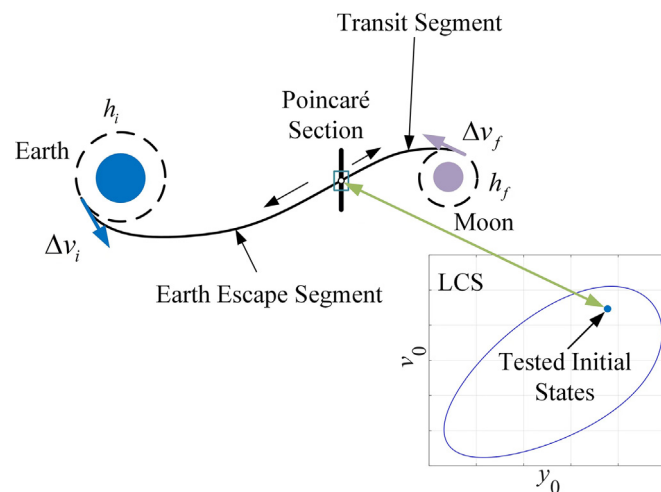


Fig. 6. LCS-based bi-impulsive Earth-Moon transfer. The LCS-based bi-impulsive Earth-Moon transfers consist of Earth escape and transit segments. The initial states of them are identical and selected from inside the LCS.

time propagation of the same initial states. The departure and arrival epochs at the Earth parking orbit and the target circumlunar orbit are denoted as t_i and t_f , respectively. Therefore, the flow of the Earth escape and transit segments can be expressed by $\phi_{t_i}^{t_0}(\mathbf{X}_0)$ and $\phi_{t_f}^{t_0}(\mathbf{X}_0)$, where $\mathbf{X}_0 = [x_0, y_0, u_0, v_0]^T$ is the initial state inside the LCS on the U_1 section, as defined by Eqs. 6,7. The trajectories are propagated using the variable step-size, variable order (VSVO) Adams–Bashforth–Moulton algorithm with absolute and relative tolerances set to 1×10^{-11} , which is performed by Matlab® 'ode113' command (Oshima, 2021). Subsequent subsections detail the construction process of LCS-based bi-impulsive transfers with low Δv_f , including selecting (θ_{s0}, H_0) parameters for LCSs, generating initial guesses, recording arrival-in performance parameters, analyzing arrival-in performance, establishing the dependence law and construction strategy, and performing the trajectory correction. In particular, interior-LCSs help to construct interior transfers, while exterior-LCSs help to construct exterior transfers.

3.2. Selection of (θ_{s0}, H_0) parameters for LCSs

To obtain the arrival-in performance of initial guesses and establish the dependence law of Δv_f , we preliminarily focus on the transit segments. As mentioned in Section 2.2 (see in Fig. 6), the transit segment generated by the LCS under the specific (θ_{s0}, H_0) parameters. In the CR3BP, Δv_f tends to decrease as the energy level reduces (Ren and Shan, 2014). Therefore, to construct the Earth-Moon transfers with low Δv_f , we investigate the LCS-based initial guesses with relatively low levels of H_0 . In the following texts, we assume that $\theta_{s0} \in [0, 360](\text{deg})$ with a stepsize of 90 deg and H_0 belongs to the interval of $[H_{0\min}, H_{0\min} + 0.04] \text{ (LU/TU)}^2$ with a stepsize of 0.005 (LU/TU)^2 . The almost invariance of the LCS implies that similar conclusions can be drawn even when selecting Poincaré sections that differ from those used in this paper. Subsequently, the initial guesses are generated and the arrival-in performance parameters are recorded.

3.3. Generation of initial guesses for bi-impulsive transfers

Both the transit and the Earth escape segments of the initial guesses are generated from the initial states inside the LCSs (both interior-LCSs and exterior-LCSs) under the (θ_{s0}, H_0) parameters selected in Section 3.2. The initial states (y_0, v_0) are searched on square grids inside the LCSs (the stepsizes of the grids are set to 1×10^{-4} for interior-LCSs and 5×10^{-5} for exterior-LCSs). The stepsize selection criterion ensures that when the stepsize is smaller than the chosen value, the qualitative conclusions remain unchanged, while optimizing the grid search time. The integration time for the transit and Earth escape segments is respectively set to 7 TU and 50 TU (i.e., $t_0 - t_i \leq 50\text{TU}$

and $t_f - t_0 \leq 7\text{TU}$). Therefore, the time of flight (TOF) of transfers investigated in this paper is limited in 57 TU. The propagation terminates when the trajectory intersects the surface of the Earth or the Moon to exclude the Earth/Moon collision orbits (Topputo, 2013).

The Earth escape segments $\phi_{t_i}^{t_0}(\mathbf{X}_0)$ and transit segments $\phi_{t_0}^{t_f}(\mathbf{X}_0)$ of the initial guesses satisfy the following conditions

$$\begin{cases} \sqrt{(x_i + \mu)^2 + y_i^2} = R_E + h'_i \in [R_E + h_i - h_{e1}, R_E + h_i + h_{e1}] \\ (x_i + \mu)(u_i - y_i) + y_i(v_i + x_i + \mu) = 0 \end{cases} \quad (15)$$

$$\begin{cases} \sqrt{(x_f + \mu - 1)^2 + y_f^2} = R_M + h'_f \in [R_M + h_f - h_{e2}, R_M + h_f + h_{e2}] \\ (x_f + \mu - 1)(u_f - y_f) + y_f(v_f + x_f + \mu - 1) = 0 \end{cases} \quad (16)$$

where h_{e1} and h_{e2} are the released altitude range considering the altitude tolerances for the Earth parking and target circumlunar orbits. Here, we set them as $h_{e1} = 1800\text{km}$, $h_{e2} = 5\text{km}$. When the grid search for each LCS is completed, states of initial guesses satisfying Eqs. 15,16 are recorded. The recorded data includes the (θ_{s0}, H_0) parameters of the LCS, initial states inside the LCS (i.e., y_0 and v_0), as well as t_i and t_f .

Then, these developed LCS-based initial guesses will be slightly corrected to rigorously satisfy the altitude constraints of the Earth parking orbit (36000 km) and the target circumlunar orbit (100 km). Before this correction, the arrival-in performance of the initial guesses help to further select the appropriate (θ_{s0}, H_0) parameters for both interior-LCSs and exterior-LCSs to construct the bi-impulsive transfers with low Δv_f .

3.4. Arrival-in performance parameters of initial guesses

In this paper, we record E_f, M_f, C_f and Δv_f at the lunar insertion point of the initial guesses to describe the arrival-in performance. E_f, M_f, C_f is given by (Oshima et al., 2019)

$$E_f = \frac{1}{2} \left[(u_f - y_f)^2 + (v_f + x_f + \mu - 1)^2 \right] - \frac{\mu}{\sqrt{(x_f + \mu - 1)^2 + y_f^2}} \quad (17)$$

$$M_f = (x_f + \mu - 1)(v_f + x_f + \mu - 1) - y_f(u_f - y_f) \quad (18)$$

$$C_f = -\left(u_f^2 + v_f^2\right) + \left(x_f^2 + y_f^2\right) + \frac{2(1 - \mu)}{\sqrt{(x_f + \mu)^2 + y_f^2}} + \frac{2\mu}{\sqrt{(x_f + \mu - 1)^2 + y_f^2}} + \mu(1 - \mu) \quad (19)$$

where $\mathbf{X}_f = [x_f, y_f, u_f, v_f]^T$ satisfies Eq. (16). If $E_f \leq 0$, the spacecraft is ballistically captured by the Moon (Topputo, 2013). Moreover, if $M_f > 0$, the spacecraft inserts into the target circumlunar orbit directly whereas $M_f < 0$ means retrograde insertion.

3.5. Analysis of arrival-in performance

This subsection presents numerical results of arrival-in performance. Our results demonstrate that the signs of all the tested E_f values are negative, indicating that all the developed trajectories can be ballistically captured by the Moon. Utilizing lunar ballistic capture (LBC) allows for an effective reduction in the Δv_f (Belbruno and Miller, 1993), a result achieved in our work through the construction of Earth-Moon transfers using LCSs. Furthermore, LBC is categorized into direct and retrograde captures based on the sign of M_f (Oshima et al., 2019). Then, the characteristics of Δv_f are discussed for interior and exterior initial guesses, direct and retrograde capture.

3.5.1. Interior initial guesses

Fig. 7 presents an example for the distribution of Δv_f and C_f for initial guesses based on interior-LCSs, which exhibit a banded pattern under the specific (θ_{s0}, H_0) parameters. This pattern is possibly attributed to the variations in h'_f , within the range $h'_f \in [h_f - h_{e2}, h_f + h_{e2}]$. Appendix A estimates the effects of h_{e2} on the difference in Δv_f to illustrate the consistency of the arrival-in performance analysis between initial guesses (i.e., trajectories satisfying Eqs. 15,16) and those satisfying Eq. (10)–(11). The distribution of Δv_f and C_f can be segmented into two main parts with different signs of M_f , i.e., direct and retrograde capture under the specific (θ_{s0}, H_0) parameters (note that the exis-

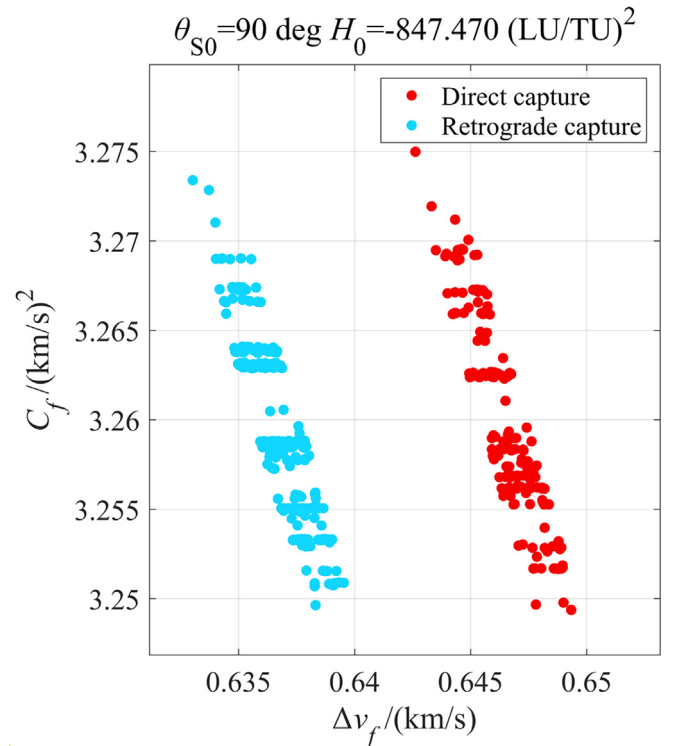


Fig. 7. An example for $(\Delta v_f, C_f)$ distributions of initial guesses. The $(\Delta v_f, C_f)$ distribution can be segmented into two main parts with direct/retrograde capture when the H_0 value is appropriate.

tence of direct and retrograde capture both requires a specific H_0 value at each θ_{S0} , which is detailed in following texts). When (θ_{S0}, H_0) parameters vary, the distribution of Δv_f and C_f also varies.

Fig. 8 presents the distribution of Δv_f and C_f for initial guesses based on interior LCS across the range of H_0 values at each θ_{S0} selected in Section 3.2 (the number in the label indicates the value of H_0). Despite the banded and overlapping distributions of Δv_f and C_f at different (θ_{S0}, H_0) , Fig. 8 clearly shows the trend of Δv_f varying with H_0 and θ_{S0} . The results show that the distribution is mainly segmented into two parts, i.e., direct and retrograde capture, which can be considered as a new phenomena compared to the results reported by Ren and Shan (2014). Specifically, direct capture is characterized by a relatively higher C_f and occurs over a broader range of H_0 values. For direct capture, the absence of data points for $H_0 = -847.515(\text{LU/TU})^2$ at $\theta_{S0} = 0, 180$ deg indicates that the spacecraft does not satisfy the Eqs. 15,16 at these θ_{S0} values. Therefore, a critical H_0 (H_{0c}) exists for both direct and retrograde capture at each θ_{S0} ; the capture is feasible only when $H_0 \geq H_{0c}$ (i.e., the necessary condition of H_0 for direct/retrograde capture). Notably, for retrograde capture, H_{0c} is higher than that required for direct capture. The SGP influences the arrival-in performance by varying H_{0c} for different θ_{S0} values for direct/retrograde capture. Interestingly, qualitative similarities in the arrival-in performance are observed at $\theta_{S0} = 0, 180$ deg and $\theta_{S0} = 90, 270$ deg, reflecting similar dynamical behavior and transfer characteristics of transit segments at two θ_{S0} values differing by 180 deg.

For the same θ_{S0} value, the maximum Δv_f is found to be positively correlated to the level of H_0 both for direct and retrograde capture. Meanwhile, a negative correlation between Δv_f and C_f has been observed. The results extend the conclusions from the CR3BP (Ren and Shan, 2014) to the PBCR4BP. Additionally, under the same level of H_0 , retrograde capture consistently exhibits lower Δv_f than direct capture, providing the data for the conclusion that retrograde capture is advantageous for lunar insertion (Oshima et al., 2019). From the aforementioned discussions, to achieve bi-impulsive LCS-based transfers with low Δv_f , the initial guesses generated from the LCSs under relatively low H_0 values at each θ_{S0} are prioritized. In addition, when direct and retrograde capture exist under the same H_0 value, the initial guesses with retrograde capture should be prioritized.

3.5.2. Exterior initial guesses

Fig. 9 presents the distribution of Δv_f and C_f for exterior initial guesses. Similar to the interior circumstances, Δv_f is found to be negatively correlated to C_f . For retrograde capture, a relatively higher level of H_{0c} is required (i.e., $-847.475(\text{LU/TU})^2$ for $\theta_{S0} = 0, 180$ deg while $-847.495(\text{LU/TU})^2$ for $\theta_{S0} = 90, 270$ deg). Anomalies in the retrograde capture at $\theta_{S0} = 0, 180$ deg are observed,

where the maximum Δv_f under $-847.465(\text{LU/TU})^2$ is unexpectedly lower than under $-847.475, -847.470(\text{LU/TU})^2$. However, the overall characteristics of arrival-in performance for the exterior circumstances align with those observed in the interior circumstances. The main differences between the interior and exterior circumstances are the different H_0 levels needed for LCS existence (see Table 2) and the H_{0c} for direct/retrograde capture. Table 3 summarizes the H_{0c} at each θ_{S0} for direct and retrograde capture in interior and exterior circumstances, revealing that the variation of with is more pronounced in the exterior circumstances. This suggests a more significant influence of the SGP in exterior circumstances compared to interior ones.

3.6. Dependence law of Δv_f and construction strategy of transfers with low Δv_f

Based on the aforementioned discussion, we have drawn the following conclusions:

- (1) All the tested LCS-based initial guesses result in lunar ballistic capture, leading to a reduction of Δv_f ;
- (2) The occurrence of direct and retrograde capture requires a minimum H_0 ($H_0 \geq H_{0c}$) at each θ_{S0} , determinable numerically;
- (3) Retrograde capture typically requires higher H_{0c} than direct capture;
- (4) Variations in θ_{S0} yield different H_{0c} , indicating the effect of SGP on the LCS-based transfers, with qualitative properties being similar for θ_{S0} differences of 180 deg;
- (5) According to Table 3, it is found that the effect of SGP on exterior transfers is more significant;
- (6) The maximum Δv_f for each θ_{S0} generally shows a positive correlation with the level of H_0 for both direct and retrograde capture;
- (7) Under the same H_0 conditions, retrograde capture results in a lower Δv_f than direct capture.

Based on these conclusions, we propose a construction strategy for low-energy transfers with low Δv_f . This strategy is used to select appropriate initial guesses. We select the initial guesses from the three lowest H_0 values at each θ_{S0} ($\theta_{S0} = 0, 90, 180, 270$ deg) for interior and exterior LCS-based transfers, and both direct and retrograde captures based on conclusions (4)-(6). This strategy filters out initial guesses which have the potential to end up with high Δv_f . Subsequently, the trajectory correction is performed to obtain bi-impulsive transfers satisfying Eqs. 10,11, which is detailed in Section 3.7.

3.7. Trajectory correction

After selecting LCS-based initial guesses as discussed in Section 3.6, the trajectory correction is performed to rigor-

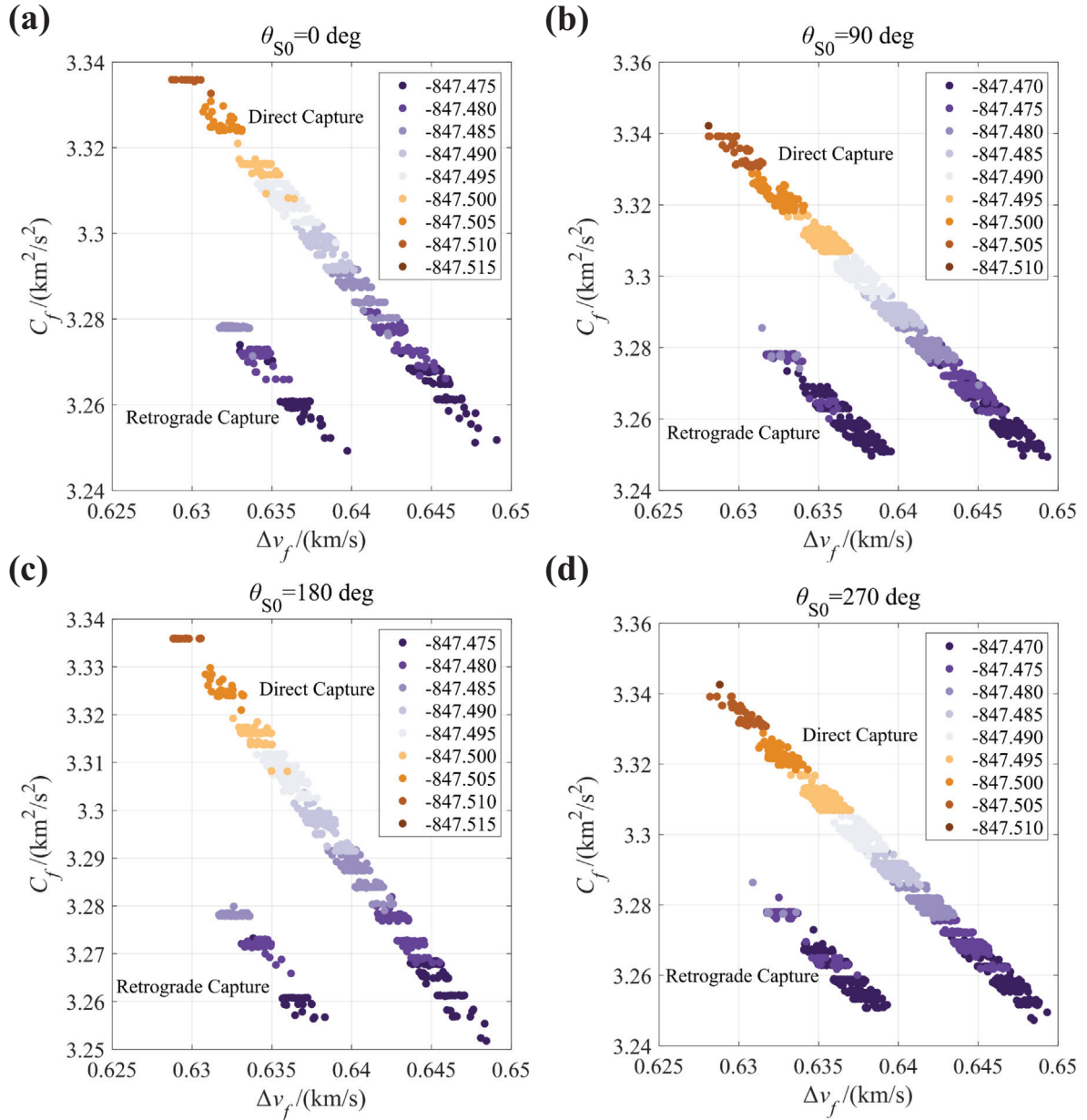


Fig. 8. $(\Delta v_f, C_f)$ distribution for interior initial guesses. (a) $\theta_{s0} = 0$ deg. (b) $\theta_{s0} = 90$ deg. (c) $\theta_{s0} = 180$ deg. (d) $\theta_{s0} = 270$ deg.

ously align the trajectories with Eqs. 10,11. The trajectory correction is achieved by solving the NLP problem (Topputo, 2013). For selected initial guesses under the specific (θ_{s0}, H_0) parameters mentioned in Section 3.6, The NLP variables are defined as $y = \{y_0, v_0, t_i, t_f\}$. To ensure the corrected trajectories satisfy Eqs. 10,11, we define the equality constraints of the NLP problem to be those constraints. This setup yields a system of four variables matched to four constraints. Additionally, the objective function of the NLP problem is set as $\left\| \begin{bmatrix} \psi_i^T \\ \psi_f^T \end{bmatrix} \right\|$. The NLP problem is solved by using the Matlab® 'fmincon' command, performing the active set method with tolerance on equality constraints of 5×10^{-8} . The ranges of

the NLP variables are set in Table 4. Since the trajectories are corrected from the LCS-based initial guesses, the method of constructing bi-impulsive transfers in this paper is denoted as LCS-based correction.

Above all, the process of constructing LCS-based bi-impulsive transfers with low Δv_f is summarized in Fig. 10. Furthermore, since LCSs are applicable to both autonomous and non-autonomous systems (Short and Howell, 2014), the aforementioned analysis of arrival-in performance and the proposed construction strategy can be extended to higher-fidelity models (e.g., the ephemeris model). The results of LCS-based transfers are presented and discussed in Section 4. To link with previous works (Battin, 1999; Topputo, 2013; Ren & Shan, 2014; Oshima

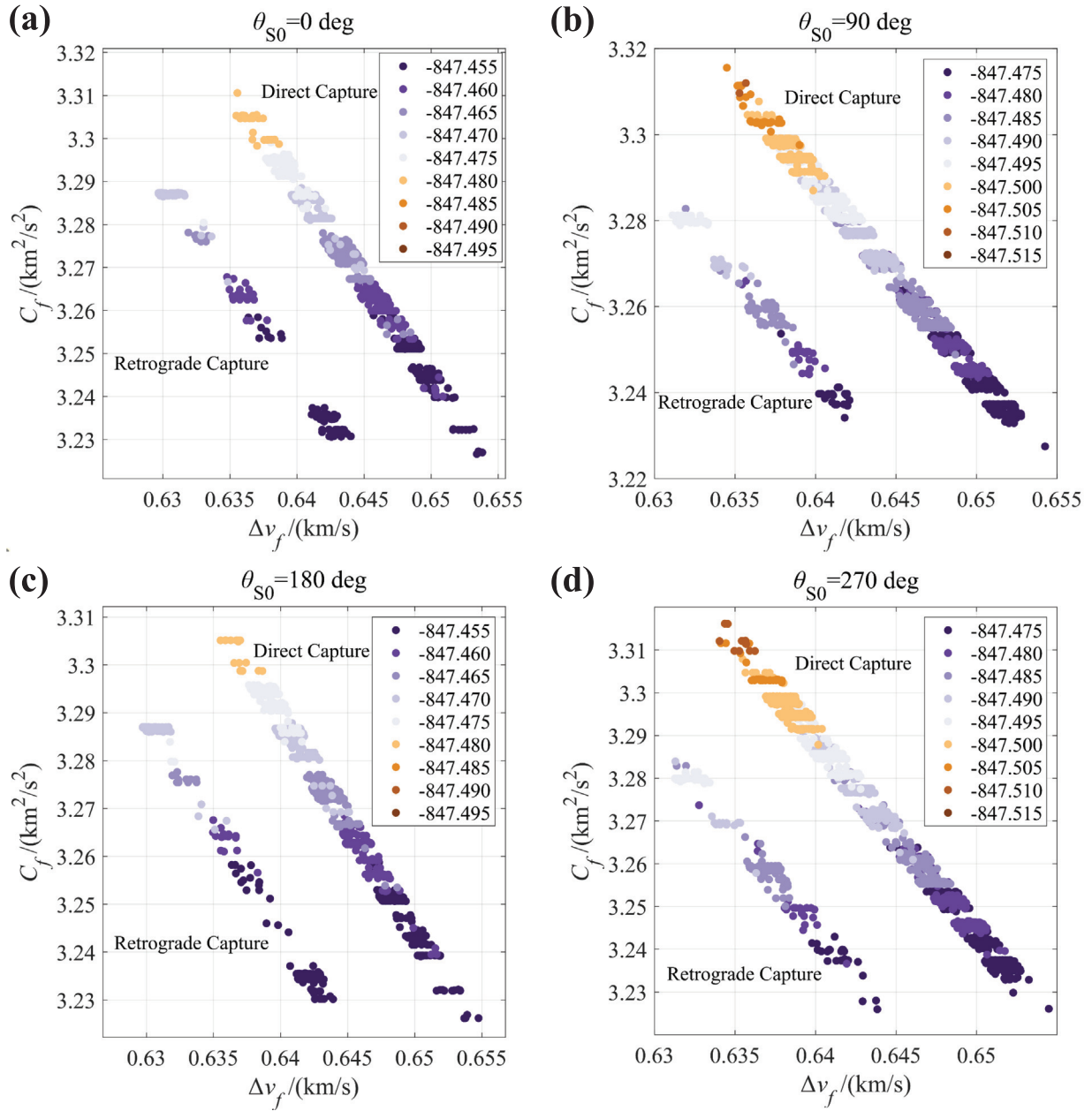


Fig. 9. $(\Delta v_f, C_f)$ distribution for exterior initial guesses. (a) $\theta_{s0} = 0$ deg. (b) $\theta_{s0} = 90$ deg. (c) $\theta_{s0} = 180$ deg. (d) $\theta_{s0} = 270$ deg.

Table 3

$H_{0c}/(LU/TU)^2$ at each θ_{s0} for four scenarios.

θ_{s0}/deg	0	90	180	270
Interior Initial Guesses, Direct Capture	-847.510	-847.510	-847.510	-847.510
Interior Initial Guesses, Retrograde Capture	-847.485	-847.480	-847.485	-847.480
Exterior Initial Guesses, Direct Capture	-847.480	-847.510	-847.480	-847.510
Exterior Initial Guesses, Retrograde Capture	-847.475	-847.495	-847.475	-847.495

Table 4

Initial guesses ranges setting in the 'fmincon' command.

NLP variables	y_0/LU	$v_0/(LU/TU)$	t_i/TU	t_f/TU
Ranges	$[-10, 10]$	$[-10, 10]$	$[t_i - 1, t_i + 1]$	$[t_f - 1, t_f + 1]$

et al., 2019), the TOF and Δv of the developed transfers are also recorded.

4. Results and discussions

This section presents the results of the LCS-based transfer trajectories. Firstly, the full data of results are summarized, including the catalog the selected LCSs and the overall TOF, Δv and Δv_f distributions of obtained trajectories. Typical transfers with minimum Δv_f and Δv for aforementioned four scenarios are detailed and discussed. Finally, a comparative analysis with transfers constructed based on other dynamical models is performed. This comparison reveals the role of SGP, LCS, and LBC in reducing Δv and Δv_f of bi-impulsive transfers.

4.1. Summarization of full data

Table 5 shows the catalog of the selected initial guesses (i.e., selected LCSs) to construct the transfers. Fig. 11 illustrates the distributions of TOF, Δv , and Δv_f for LCS-based transfers alongside the optimization solutions in the PBCR4BP. The optimization results, represented by orange dots, are obtained using the boundary value problem (BVP) and NLP approach with direct optimization, detailed in Appendix B (Topputo, 2013). The cyan stars represent the developed LCS-based solutions for the four scenarios. This comparison reveals that, across these scenarios, the LCS-based construction strategy yields relatively low lunar insertion impulses (0.6282–0.6444 km/s) without direct optimization, showing an advantage over the optimization results. This benefit is attributed to the incorporation of LBC and the utilization of the natural dynamical structure LCS, verifying the effectiveness of our proposed strategy.

Additionally, a significant number of developed LCS-based transfers are characterized by relatively low Δv (1.6148–1.700 km/s) and long TOF (100–243 days) compared to the optimization solutions (orange dots). Therefore,

these findings suggest that the majority of developed LCS-based solutions qualify as low-energy transfers. Consequently, low-energy transfers in the PBCR4BP are constructed in a more targeted way, without the need for a global search as required by direct optimization. This can be considered as an advantage of the LCS-based strategy. Transfers exhibiting higher Δv typically follow 'exterior-interior' patterns or are purely exterior, requiring relatively high Δv_i to initiate departure from Earth parking orbit without effective lunar gravity assist (LGA) (Oshima et al., 2019). Some samples of these trajectories are presented in Appendix C. Table 6 summarizes the TOF, Δv and Δv_f ranges of developed transfers. In particular, our findings indicate that low-energy transfers are mainly associated with interior LCS-based transfers, especially in direct capture, while exterior LCS-based transfers exhibit a broader TOF range. For constructing low-energy transfers with low Δv_f , the interior LCS-based transfers with direct capture are prioritized. For interior LCS-based transfers, though retrograde capture is advantageous for lunar insertion, the Δv_f of retrograde capture tends to be higher than for direct capture due to the significantly higher H_{0c} (see conclusion (6) in Section 3.6). Subsequently, samples of the developed transfers with minimum Δv and Δv_f for four scenarios are investigated and discussed.

4.2. Samples of interior LCS-based transfers with direct capture

Two new interior transfers (i.e., sample I and sample II) are presented in Fig. 12 and Fig. 13. The orange stars on the trajectories indicate the initial states inside the LCSs. Sample I and sample II represent the minimum Δv (1.6148 km/s) and Δv_f (0.6282 km/s) solutions for interior LCS-based transfer with direct capture, respectively. The TOFs of sample I and sample II are 139 and 205 days, respectively. The transit segments in these samples illustrate a tour trajectory (Canales et al., 2023). Fig. 12 (b)

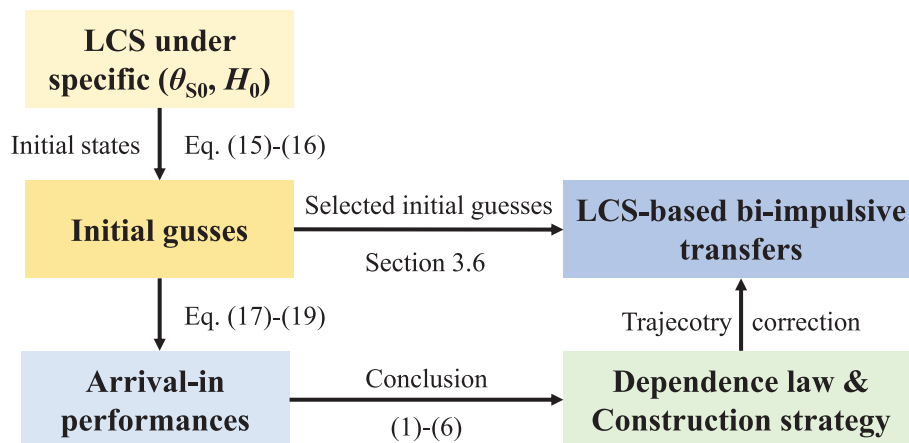


Fig. 10. Process of constructing the LCS-based bi-impulsive transfers.

Table 5
Catalog of the selected initial guesses.

θ_{S0}/deg	0	90	180	270
Interior Initial Guesses, Direct Capture	–847.510	–847.510	–847.510	–847.510
	–847.505	–847.505	–847.505	–847.505
	–847.500	–847.500	–847.500	–847.500
Interior Initial Guesses, Retrograde Capture	–847.485	–847.480	–847.485	–847.480
	–847.480	–847.475	–847.480	–847.475
	–847.475	–847.470	–847.475	–847.470
Exterior Initial Guesses, Direct Capture	–847.480	–847.510	–847.480	–847.510
	–847.475	–847.505	–847.475	–847.505
	–847.470	–847.500	–847.470	–847.500
Exterior Initial Guesses, Retrograde Capture	–847.475	–847.495	–847.475	–847.495
	–847.470	–847.490	–847.470	–847.490
	–847.465	–847.485	–847.465	–847.485

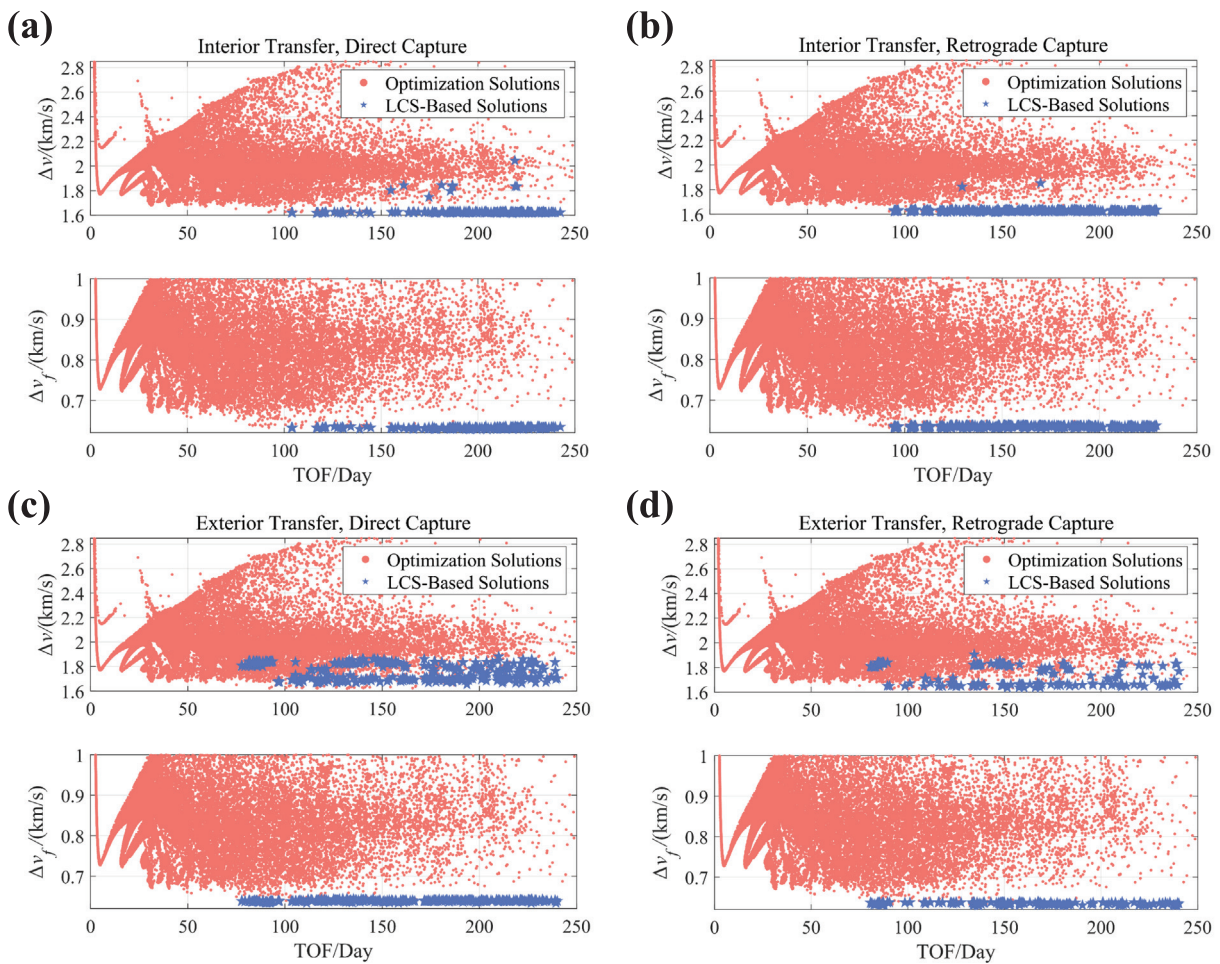


Fig. 11. Plane of TOF, Δv and Δv_f for optimization solutions and the LCS-based solutions. (a) Results of interior transfers with direct capture. (b) Results of interior transfers with retrograde capture. (c) Results of exterior transfers with direct capture. (d) Results of exterior transfers with retrograde capture.

illustrates the evolution of the semimajor axis about the Earth (a_E) in the initial part of the transfer. In the time history, a_E is pumped up by the high-altitude lunar flybys (Oshima et al., 2019) before encountering the Moon according to the fluctuation of a_E . This mechanism leads to large trajectory changes at zero cost (Koon et al., 2006) and results in low Δv_i and Δv solutions for the developed LCS-based transfers.

4.3. Samples of interior LCS-based transfers with retrograde capture

Sample III is not only the minimum Δv solution (1.6180 km/s) but also the minimum Δv_f solution (0.6326 km/s) for interior LCS-based transfer with retrograde capture. The TOF of sample III is 148 days. Fig. 14 presents the associated trajectory with a new pat-

Table 6
Range of TOF, Δv and Δv_f for four scenarios.

Circumstances	TOF/Day	Δv /(km/s)	Δv_f /(km/s)
Interior Transfers, Direct Capture	104–243	1.615–2.044	0.628–0.636
Interior Transfers, Retrograde Capture	95–229	1.618–1.851	0.633–0.639
Exterior Transfers, Direct Capture	78–240	1.655–1.882	0.634–0.644
Exterior Transfers, Retrograde Capture	81–240	1.635–1.904	0.631–0.640

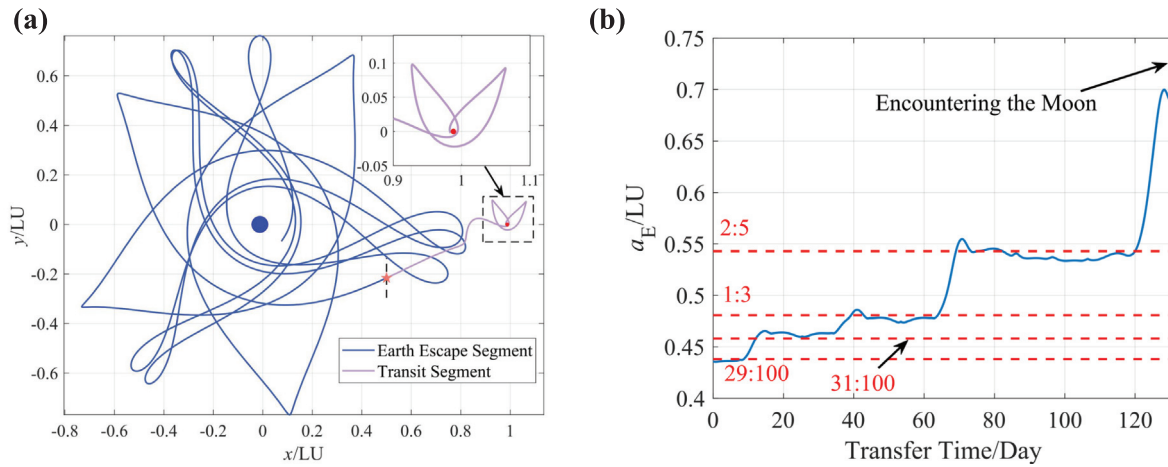


Fig. 12. Sample I. (a) Trajectory (Earth escape segment and transit segment) of sample I. (b) Variation in a_E and resonance evolution in the initial part of the transfer.

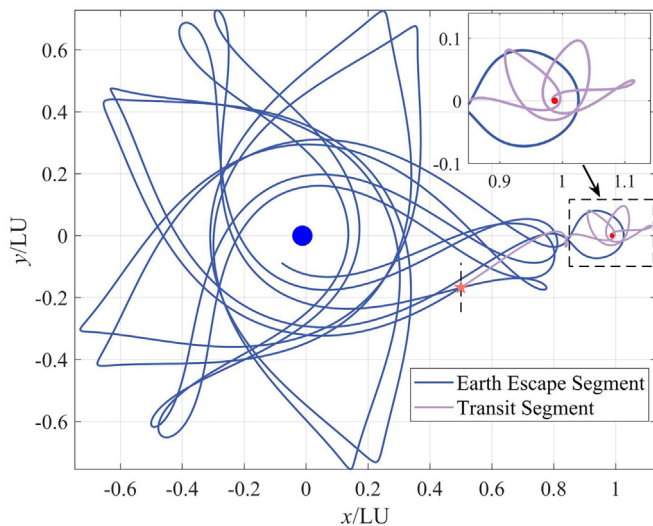


Fig. 13. Trajectory (Earth escape segment and transit segment) of sample II.

tern of transit segment and the variations in the angular momentum with respect to the Moon (M_M) during the final part of the transfer. The observed behavior of M_M in sample III closely aligns with the findings from Oshima et al. (2019) for exterior transfers. Therefore, the mechanism of retrograde capture (Oshima et al., 2019) that the bounce (see the pink curve in Fig. 14 (a)) reduces the angular

momentum with respect to the Moon is effective in both interior and exterior transfers.

4.4. Samples of exterior LCS-based transfers with direct capture

Sample IV and sample V are the minimum Δv (1.6550 km/s) and Δv_f (0.6340 km/s) solution for exterior LCS-based transfer with direct capture, respectively. The TOFs of sample IV and sample V are 194 and 111 days, respectively. In Fig. 15 (a), the trajectory of sample IV is classified as an 'interior-exterior' transfer (Oshima et al., 2019), a new low-energy pattern. This trajectory, passing through the L2 region finally and being generated from the exterior-LCS, is classified as an exterior LCS-based transfer. The dynamics of the 'interior-exterior' transfer, characterized by high-altitude lunar flybys combined with LGA in the Earth escape segments initiated from exterior-LCSs. As detailed in Section 3.5.2, these exterior-LCSs contribute to achieving a low Δv_f for the 'interior-exterior' transfer. Meanwhile, the jumping between resonance regions and LGA collectively contributes to a reduction in both the Δv and Δv_f of the transfer. Therefore, compared to direct optimization, using the LCSs to construct transfers help explain about the generation of some types of low-energy transfers from a geometric perspective. (Figs. 16–18).

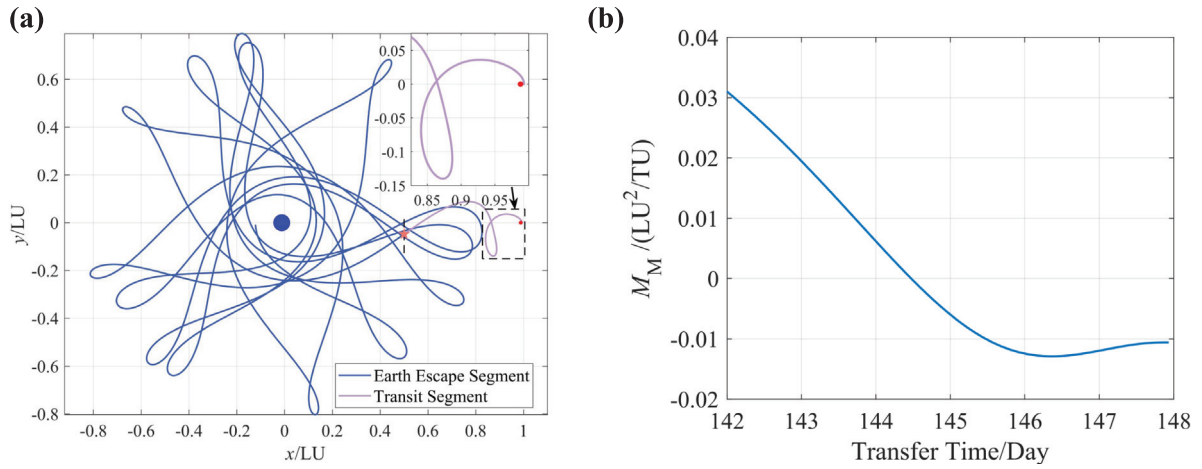


Fig. 14. Sample III. (a) Trajectory (Earth escape segment and transit segment) of sample III. (b) Variation in M_M in the final part of the transfer.

4.5. Samples of exterior LCS-based transfers with retrograde capture

Sample VI and sample VII are the minimum Δv (1.6352 km/s) and Δv_f (0.6306 km/s) solutions for exterior LCS-based transfer with retrograde capture. The TOFs of sample VI and sample VII are 152 and 237 days, respectively. These trajectories exhibit new 'interior-exterior' transfer patterns, involving high-altitude lunar flybys and LGA. Notably, the transit segment of sample VII exhibits a trajectory pattern similar to that of sample VI. The difference of Δv_f between these samples is mainly caused by different (θ_{s0}, H_0) parameters (i.e., (90 deg, -847.485 $(LU/TU)^2$) for sample VI and (0 deg, -847.470

$(LU/TU)^2$) for sample VII). This finding highlights the effectiveness of the strategy proposed in Section 3, which involves selecting a full range of θ_{s0} values.

4.6. Comparison with results in other dynamical models

To clarify the helpful role of the SGP and the LCS in reducing Δv_f and Δv , we compare the aforementioned seven LCS-based samples developed by the proposed strategy with solutions in other dynamical models. The theoretical minimum Δv_f and Δv are estimated by the method proposed by Sweetser (1991) (note that this theoretical minimum impulse cannot be achieved by a transfer with a finite TOF). Solution I indicates the result reproduced

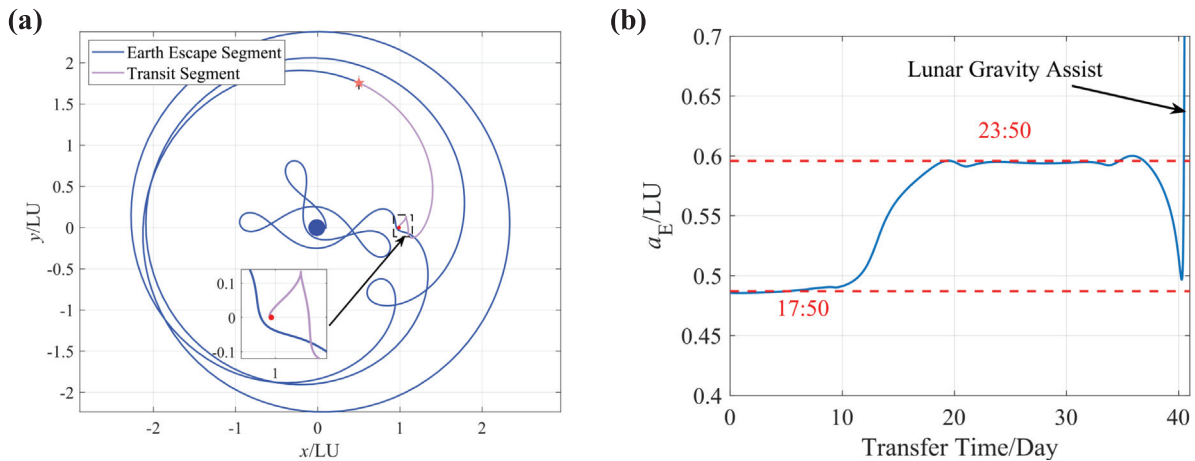


Fig. 15. Sample IV. (a) Trajectory (Earth escape segment and transit segment) of sample I. (b) Variation in a_E and resonance evolution in the initial part of the transfer.

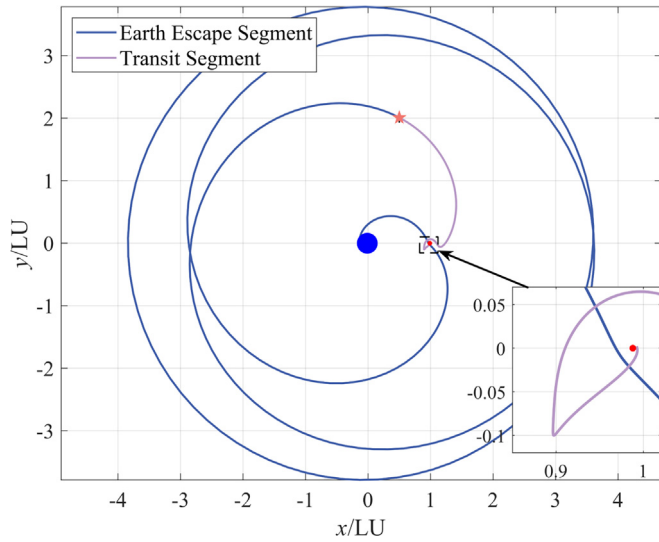


Fig. 16. Trajectory (Earth escape segment and transit segment) of sample V.

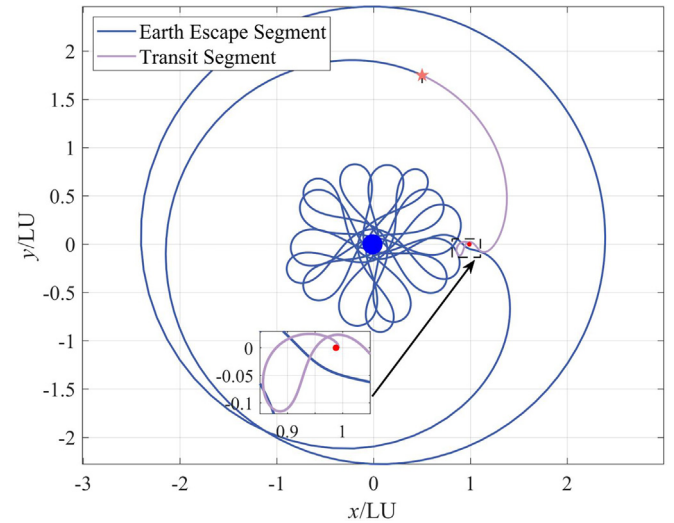


Fig. 18. Trajectory (Earth escape segment and transit segment) of sample VII.

by the patched conic method (Battin, 1999). Moreover, solution II is our reproduced minimum Δv_f and Δv solution using the BVP-NLP approach with direct optimization (Topputo, 2013; Oshima et al., 2019) in the Earth-Moon planar CR3BP (PCR3BP).

According to Table 7, it is observed that the Δv_f of samples I-VII are all close to the theoretical minimum Δv_f (0.6250 m/s) of Sweetser (1991). In comparison between the LCS-based samples and solution I, it is found that the LCS-based method combined with LBC yields an improvement in Δv and Δv_f by 2.4–8.3% and 9.9–11.5%, respectively. Furthermore, when comparing the minimum Δv and Δv_f solutions of seven samples with solution II,

the effect of the SGP accumulates and results in a slight reduction in Δv (about 0.4%) and Δv_f (about 0.9%).

5. Conclusion

This paper focuses on the lunar ballistic capture (LBC) and bi-impulsive Earth-Moon transfers with low Moon insertion impulse constructed based on the Lagrangian coherent structures (LCSs) in the Sun-Earth/Moon planar bicircular restricted four-body problem (PBCR4BP). Arrival-in performance of the trajectories have revealed the dependence law of LBC and the Moon insertion impulse. Our results indicate that the LCS-based transfers

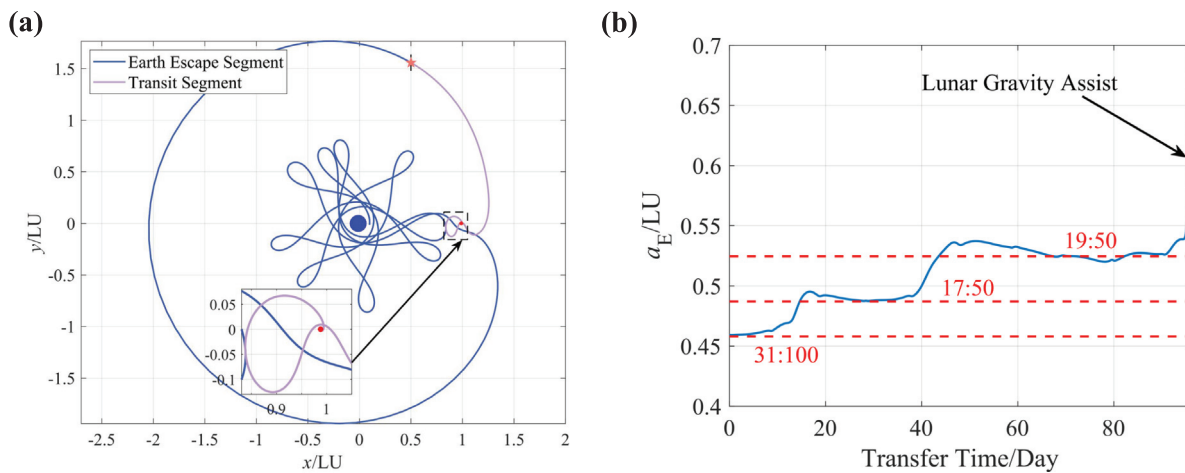


Fig. 17. Sample VI. (a) Trajectory (Earth escape segment and transit segment) of sample I. (b) Variation in a_E and resonance evolution in the initial part of the transfer.

Table 7

Comparison between developed samples and solutions under other dynamical models.

Solutions	Δv_f /(km/s)	Δv /(km/s)	h_i /km	h_f /km	Dynamical Model	Method	Type
Sample I	0.630	1.615	36000	100	PBCR4BP	LCS-based Correction	Interior
Sample II	0.628	1.620	36000	100	PBCR4BP	LCS-based Correction	Interior
Sample III	0.633	1.618	36000	100	PBCR4BP	LCS-based Correction	Interior
Sample IV	0.640	1.655	36000	100	PBCR4BP	LCS-based Correction	Exterior
Sample V	0.634	1.719	36000	100	PBCR4BP	LCS-based Correction	Exterior
Sample VI	0.635	1.635	36000	100	PBCR4BP	LCS-based Correction	Exterior
Sample VII	0.631	1.666	36000	100	PBCR4BP	LCS-based Correction	Exterior
Minimum	0.625	1.580	36000	100	PCR3BP	Theoretical Estimation	Interior
Solution I	0.709	1.761	36000	100	Patched R2BP	Patched Conic	Interior
Solution II	0.634	1.620	36000	100	PCR3BP	Direct Optimization	Interior

tend to end up in LBC, which can be categorized into direct and retrograde capture based on the sign of the angular momentum with respect to the Moon. In addition, the Moon insertion impulse generally shows a positive correlation with the energy level and is affected by the direct/retrograde capture. Sun-perturbed dynamics affects the Moon insertion impulse by influencing energy levels. Based on these laws, the construction strategy of LCS-based bi-impulsive transfers with low Moon insertion impulse for four scenarios (i.e., interior/exterior transfers with direct/retrograde capture) is proposed. Numerical simulations show that a bi-impulsive transfer with a low Moon insertion impulse (0.6282–0.6444 km/s) can be effectively achieved using the LCS and LBC. The LCS-based solutions require lower total and Moon insertion impulses compared to the two-body and three-body problems solutions, confirming the utility of the solar gravity perturbation for constructing low-energy transfers. Our contribution further establishes a link between impulses, energy levels, and Sun-perturbed dynamics based on the LCSs, and achieves a more targeted way of constructing low-energy transfers. Future research will focus on the transit orbits in the PBCR4BP and further explore the relationships between these trajectories and transfer-related parameters, such as time of flight (TOF), perilune radius, and Moon insertion impulse.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Estimation of the effect of h_{e2} on the v_f

To assess the effect of $h_{e2} = 5\text{km}$ and h'_f on the values of Δv_f , the method of compensation based on the Kepler energy with respect to the Moon E_f is adopted (Oshima et al., 2019). The E_f of the lunar insertion point with the altitude of h'_f is calculated by

$$E_f = \frac{1}{2} \left(\Delta v_f + \sqrt{\frac{\mu}{R_M + h'_f}} \right)^2 - \frac{\mu}{R_M + h'_f} \quad (20)$$

where $h'_f \in [95, 105](\text{km})$. Therefore, the required Δv_f (Δv_{fr}) can be calculated by

$$\Delta v_{fr} = \sqrt{2 \left(E_f + \frac{\mu}{R_M + h_f} \right)} - \sqrt{\frac{\mu}{R_M + h_f}} \quad (21)$$

where $h_f = 100\text{km}$. The difference between Δv_f presented in Section 3 and Δv_{fr} is estimated by

$$\varepsilon = \Delta v_f - \Delta v_{fr} \quad (22)$$

$$= \Delta v_f - \sqrt{2 \left(\left(\frac{1}{2} \left(\Delta v_f + \sqrt{\frac{\mu}{R_M + h'_f}} \right)^2 - \frac{\mu}{R_M + h'_f} \right) + \frac{\mu}{R_M + h_f} \right)} + \sqrt{\frac{\mu}{R_M + h_f}} = \varepsilon(h'_f, \Delta v_f)$$

Based on the results shown in Fig. 8 and Fig. 9, we estimate that $\Delta v_f \approx 0.6350\text{km/s}$. As a result, this leads to an estimation of $|\varepsilon_{\max}| \approx 0.0010\text{km/s}$, as shown in Fig. 19 (a). The consistency of these analytical estimations and numerical results are demonstrated by an example (interior initial guesses) presented in Fig. 19 (b).

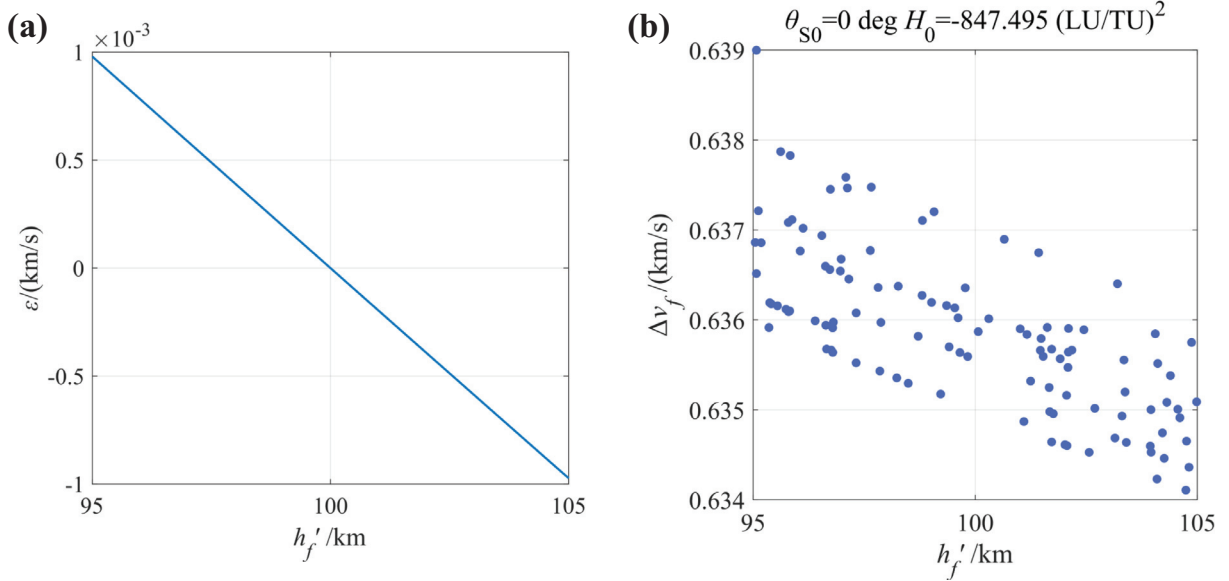


Fig. 19. Estimation of the effect of h_{e2} on the Δv_f . (a) Variation in ε with h'_f . (b) An example of numerical results.

Appendix B. Details of BVP-NLP Approach with Direct Optimization in the PBCR4BP

Similar to the construction method mentioned in Section 3, the BVP-NLP approach with direct optimization also involves generating initial guesses and performing the trajectory correction. In this appendix, we define a bi-impulsive Earth-Moon transfer using the set $\{X_i, t_i, t_f\}$. Therefore, such a bi-impulsive transfer can be expressed as $\phi_{t_i}^{t_f}(X_i)$. Satisfying the geometric constraint of the Earth parking orbit (i.e., Eq. (10)), the orbital states X_i, t_i and t_f are parameterized by (Topputo, 2013)

$$x_i = (R_E + h_i) \cos \alpha_i - \mu \quad (23)$$

$$y_i = (R_E + h_i) \sin \alpha_i \quad (24)$$

$$u_i = -\left(\beta_i \sqrt{\frac{1-\mu}{R_E + h_i}} - (R_E + h_i)\right) \sin \alpha_i \quad (25)$$

$$v_i = \left(\beta_i \sqrt{\frac{1-\mu}{R_E + h_i}} - (R_E + h_i)\right) \cos \alpha_i \quad (26)$$

$$t_i = \theta_{Si}/\omega_S \quad (27)$$

$$t_f = t_i + \text{TOF} \quad (28)$$

where α_i denotes the phase angle of the Earth parking orbit, β_i denotes the initial-to-circular velocity ratio and θ_{Si} denotes the departure epoch from the Earth parking orbit. The range and selected stepsize of $\alpha_i, \beta_i, \theta_{Si}$, and TOF (shown in Table 8) are set and transformed into

$\{X_i, t_i, t_f\}$ to generate the initial guesses. Then, the optimization problem can be transformed into a NLP problem, where the set $\{X_i, t_i, t_f\}$ serves as the optimization variables. The objective function of the NLP problem is $\Delta v = \Delta v_i + \Delta v_f$ and the constraints of the NLP problem are set as Eqs. 10,11. When an optimization solution is found, a continuation method is performed (Topputo, 2013; Oshima et al., 2019). In this continuation problem, the initial guesses are determined by the set $\{X_i, t_i, t_f, \delta\}$, where δ denotes a slight difference in TOF (see Table 8). Therefore, renewed t_f is expressed as

$$t_f = t_i + \text{TOF} + \delta \quad (29)$$

Subsequently, a new NLP problem is constructed and the direct optimization is performed again. Moreover, the optimization variables of the new NLP problem are set as $\{X_i, t_i, t_f\}$, the objective function of the NLP problem is Δv and the constraints of the NLP problem are set as Eqs. 10,11. The solutions of aforementioned two NLP problems are presented as orange dots shown in Fig. 11.

Table 8. Parameter setting in the BVP-NLP approach.

Parameter	Interval	Stepsize
α_i	$[0, 360](\text{deg})$	18 deg
β_i	$[1.00, 1.44]$	0.04
θ_{Si}	$[0, 360](\text{deg})$	18 deg
TOF	$[0.5, 50.5](\text{TU})$	2 TU
δ	$[-3, 3](\text{TU})$	0.2 TU

Appendix C. Samples of LCS-Based Transfers with High ν

Fig. 20 presents some developed trajectories with high Δv for both interior LCS-based transfers and exterior LCS-based transfers with direct/retrograde capture, respectively. Specifically, the trajectories of Fig. 20 (a) and (b) are ‘exterior-interior’ transfers. These trajectories, passing through the L1 region finally and being generated from the interior-LCSs, are classified as interior LCS-based transfers. On the other hand, the trajectories of Fig. 20 (c) and (d) are exterior transfers both with direct/retrograde capture. However, these trajectories lack an effective LGA during the transfer, resulting in a relatively high Δv_i

to enter the Earth-Moon exterior region. This requirement possibly contributes to the relatively high Δv observed. The TOF, Δv and Δv_f are presented in Table 9.

Table 9. TOF, Δv and Δv_f of samples with high Δv .

Sample	Δv /(km/s)	Δv_f /(km/s)	TOF/Day
(a)	1.8338	0.6317	220
(b)	1.8227	0.6372	129
(c)	1.8317	0.6351	154
(d)	1.8392	0.6370	87

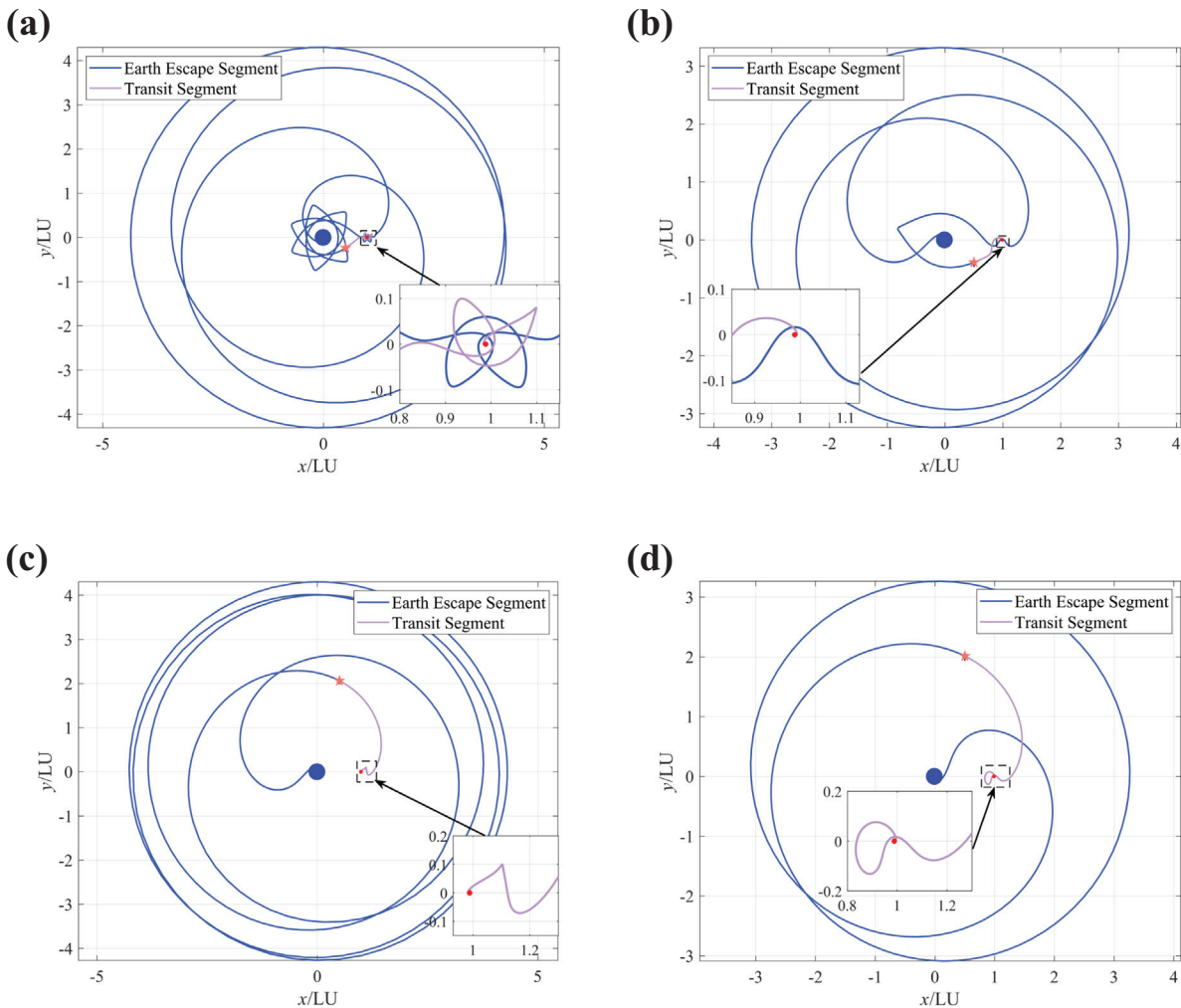


Fig. 20. Samples of LCS-based transfers with high Δv . (a) Sample of interior transfers with direct capture. (b) Sample of interior transfers with retrograde capture. (c) Sample of exterior transfers with direct capture. (d) Sample of exterior transfers with retrograde capture.

References

- Batcha, A.L., Williams, J., Dawn, T.F., et al., 2020. Artemis i trajectory design and optimization. In: 2020 AAS/AIAA Astrodynamics Specialist Conference AAS, pp. 20–649.
- Battin, R.H., 1999. An introduction to the mathematics and methods of astrodynamics. AIAA. <https://doi.org/10.2514/4.861543>.
- Belbruno, E.A., Miller, J.K., 1993. Sun-perturbed earth-to-moon transfers with ballistic capture. *J. Guid. Control Dyn.* 16 (4), 770–775. <https://doi.org/10.2514/3.21079>.
- Campagnola, S., Russell, R.P., 2010. Endgame problem part 2: multibody technique and the tisserand-poincare graph. *J. Guid. Control Dyn.* 33 (2), 476–486. <https://doi.org/10.2514/1.44290>.
- Canales, D., Howell, K.C., Fantino, E., et al., 2023. Transfers between moons with escape and capture patterns via lyapunov exponent maps. *J. Guid. Control Dyn.* 46 (11), 2133–2149. <https://doi.org/10.2514/1.G007195>.
- Cardona, T., Seitzer, P., Rossi, A., et al., 2016. Bvri photometric observations and light-curve analysis of geo objects. *Adv. Space Res.* 58 (4), 514–527. <https://doi.org/10.1016/j.asr.2016.05.025>.
- Crenshaw, J.W., 1963. Sphere of influence in patched-conic methods. *AIAA J.* 1 (9), 2168–2170. <https://doi.org/10.2514/3.2024>.
- Du, C., Song, L., Zhang, J., et al., 2024. A novel calculation method for low-thrust transfer trajectories in the earth-moon restricted three-body problem. *Aerosp. Sci. Technol.* 147, 109048. <https://doi.org/10.1016/j.ast.2024.109048>.
- Du, C., Wu, K., Starinova, O.L., et al., 2023. Two trajectory configurations for the low-thrust transfer between northern and southern halo orbits in the earth-moon system. *Adv. Space Res.* 72 (10), 4093–4105. <https://doi.org/10.1016/j.asr.2023.08.007>.
- Fu, S., Wu, D., Gong, S., 2024. Interior transit orbits in the planar bicircular restricted four-body problem: classification and application. *arXiv, astro-ph.EP*. doi:<https://doi.org/10.48550/arXiv.2407.03871>.
- Gong, S., Li, J., Baoyin, H., et al., 2007. Lunar landing trajectory design based on invariant manifold. *Appl. Math. Mech.-Engl. Ed.* 28 (2), 201–207.
- Hatch, S., Roncoli, R., Sweetser, T., 2010. Grail trajectory design: lunar orbit insertion through science. In: AIAA/AAS Astrodynamics Specialist Conference, p. 8385.
- Koon, W.S., Lo, M.W., Marsden, J.E., et al., 2001. Low energy transfer to the moon. *Celest. Mech. Dyn. Astron.* 81 (1–2), 63–73. <https://doi.org/10.1023/A:1013359120468>.
- Koon, W.S., Lo, M.W., Marsden, J.E., et al., 2006. Dynamical systems, the three-body problem and space mission design. *World Scientific*.
- Liu, C., Gong, S., 2018. Hill stability of the satellite in the elliptic restricted four-body problem. *Astrophys. Space Sci.* 363 (8), 162. <https://doi.org/10.1007/s10509-018-3381-3>.
- Mingotti, G., Topputo, F., Bernelli-Zazzera, F., 2012. Efficient invariant-manifold, low-thrust planar trajectories to the moon. *Commun. Nonlinear Sci. Numer. Simul.* 17 (2), 817–831. <https://doi.org/10.1016/j.cnsns.2011.06.033>.
- Onozaki, K., Yoshimura, H., 2013. Lunar capture trajectories in the four-body problem. In: AIAA SPACE 2013 Conference and Exposition, p. 5425.
- Onozaki, K., Yoshimura, H., Ross, S.D., 2017. Tube dynamics and low energy earth-moon transfers in the 4-body system. *Adv. Space Res.* 60 (10), 2117–2132. <https://doi.org/10.1016/j.asr.2017.07.046>.
- Oshima, K., 2021. Capture and escape analyses on planar retrograde periodic orbit around the earth. *Adv. Space Res.* 68 (9), 3891–3902.
- Oshima, K., Topputo, F., Yanao, T., 2019. Low-energy transfers to the moon with long transfer time. *Celest. Mech. Dyn. Astron.* 131, 1–19. <https://doi.org/10.1007/s10569-019-9883-7>.
- Pan, S., Topputo, F., Hou, X., et al., 2022. On optimal three-impulse earth-moon transfers in a four-body model. In: International astrodynamics congress: IAC Proceedings. International Astronautical Federation, IAF, pp. 1–9.
- Parker, J.S., Anderson, R.L., 2014. Low-energy lunar trajectory design, volume 12. John Wiley & Sons.
- Pernicka, H., Scarberry, D., Marsh, S., et al., 1994. A search for low delta-v earth-to-moon trajectories. In: Astrodynamics Conference, p. 3772.
- Pollpeter, K., Barrett, E., Edmonds, J., et al., 2023. China-Russia space cooperation: The strategic, military, diplomatic, and economic implications of a growing relationship. China Aerospace Studies Institute.
- Qi, R., Xu, S., Zhang, Y., et al., 2012. Earth-to-moon low energy transfer using time-dependent invariant manifolds. In: AIAA/AAS Astrodynamics Specialist Conference, p. 4431.
- Ren, Y., Shan, J., 2014. Low-energy lunar transfers using spatial transit orbits. *Commun. Nonlinear Sci. Numer. Simul.* 19 (3), 554–569. <https://doi.org/10.1016/j.cnsns.2013.07.020>.
- Renju, R., Raju, C.S., 2023. Simulation analysis of microwave emission from lunar subsurface for sar radiometric mode dual frequency (l/s bands) observations onboard chandrayaan-2. *Adv. Space Res.* 72 (8), 3499–3508. <https://doi.org/10.1016/j.asr.2023.06.040>.
- Short, C.R., Howell, K.C., 2014. Lagrangian coherent structures in various map representations for application to multi-body gravitational regimes. *Acta Astronaut.* 94 (2), 592–607. <https://doi.org/10.1016/j.actaastro.2013.08.020>.
- Singh, S.K., Taheri, E., Woollands, R., et al., 2019. Mission design for close-range lunar mapping by quasi-frozen orbits. In: In 70th International Astronautical Congress, Washington DC, USA.
- Song, Y.-J., Bang, J., Bae, J., et al., 2023. Lunar orbit acquisition of the korea pathfinder lunar orbiter: design reference vs actual flight results. *Acta Astronaut.* 213, 336–343. <https://doi.org/10.1016/j.actaastro.2023.09.021>.
- Sweetser, T., 1991. An estimate of the global minimum dv needed for earth-moon transfer. In: In the Spaceflight mechanics 1991, pp. 111–120.
- Topputo, F., 2013. On optimal two-impulse earth-moon transfers in a four-body model. *Celest. Mech. Dyn. Astron.* 117, 279–313. <https://doi.org/10.1007/s10569-013-9513-8>.
- Yagasaki, K., 2004. Sun-perturbed earth-to-moon transfers with low energy and moderate flight time. *Celest. Mech. Dyn. Astron.* 90, 197–212. <https://doi.org/10.1007/s10569-004-0406-8>.
- Yamato, H., Spencer, D.B., 2004. Transit-orbit search for planar restricted three-body problems with perturbations. *J. Guid. Control Dyn.* 27 (6), 1035–1045. <https://doi.org/10.2514/1.4524>.
- Zheng, Y., Mengfei, Y., Xiangjin, D., et al., 2023. Analysis of chang'e-5 lunar core drilling process. *Chin. J. Aeronaut.* 36 (2), 292–303. <https://doi.org/10.1016/j.cja.2022.01.023>.